shop desk

Complete product & engineering blueprint

MONGODB + JAVASCRIPT / VERCEL + ATLAS

Full overview • operating instructions • architecture • build sequence • legal gates • acceptance tests • copy-ready prompt

Approved target

• Independent-clinic multi-tenant SaaS

• Billing and inventory, customers/pets/suppliers; no clinical or prescription-authoring module

• Full double-entry accounting retained by express user approval

• Empty production business data; readable UI, immutable history, original-batch returns and separate refunds

Documentation, not a new application

No new SaaS was implemented, deployed or runtime-tested by preparing this pack. Professional legal/tax/licensing review and release evidence remain required. No document guarantees universal Indian-law compliance.

Client/tenant brand: The Good Vets
Product name: shop desk
Prepared 26 October 2026 · Specification edition 1.0

Reading guide

Use the navigation pane/headings in Word, or read chapter by chapter. Use the standalone Markdown/TXT prompt when pasting into a coding agent; the Word edition is the combined reading copy.

01 Start here: approved scope and how to use this pack

02 Product overview and operating guide

03 Architecture and step-by-step build

04 India legal and release matrix

05 Acceptance tests and attachment traceability

06 Vercel + Atlas online operations

07 Copy-ready master implementation prompt

08 Official and technical source register

The complete unmodified user legal notes are included separately in the reference folder. The legal matrix distinguishes observed sources, applicability questions, future commencement and research limits.

01 Start here: approved scope and how to use this pack

Client/tenant brand: The Good Vets · Prepared 26 October 2026 · Specification edition 1.0

This is a product and engineering specification, operating guide and copy-ready AI build prompt. It is **not a newly built application, a deployment, a GST certification or a legal opinion**. No hosting account, database, subscription or paid resource was created for this pack.

Your approved decisions

- Build an online **multi-tenant SaaS** for independent veterinary clinics and related shops.
- Use **JavaScript/ES modules, React/Vite, a Node.js API on Vercel, and MongoDB Atlas**. Do not introduce a separate always-running backend without approval.
- Clinical scope is **billing and inventory only**: customers, pets and suppliers are included, but diagnosis, treatment records and prescription authoring are not.
- **Retain full double-entry accounting** as your expressly approved extension to section 14 of the attached legal notes. Other exclusions remain in effect.
- Keep the software name **shop desk**. “The Good Vets” is the client/tenant brand, not an assumed registered legal entity. Statutory documents must use the actual verified supplier identity.
- Preserve returns, immutable historical money, named cashiers, reports, equal applicable new intra-state tax components, scrolling navigation, readable UI, invoice templates and thermal printing.
- New production tenants start empty. Only approved system references and a chart of accounts may be initialized; never create demo business records.

Files and reading order

File	Purpose
`PRODUCT_OVERVIEW_AND_USER_GUIDE.md`	What every module does, where it appears, who can use it, and daily operating procedures.
`ARCHITECTURE_AND_BUILD_GUIDE.md`	MongoDB model, tenant isolation, JavaScript service boundaries, APIs, exact money, transactions, job design and phased construction.
`INDIA_LEGAL_AND_RELEASE_MATRIX.md`	Official-source legal research, applicability caveats, engineering controls, unverified items and professional release gates.
`TEST_ACCEPTANCE_AND_TRACEABILITY.md`	Acceptance scenarios and mapping of all 20 sections of your legal attachment.
`VERCEL_ATLAS_OPERATIONS.md`	Online setup, private configuration, Atlas connectivity/TLS, migrations, monitoring, backups and incident recovery.
`MASTER_IMPLEMENTATION_PROMPT.md`	Standalone, copy-ready engineering prompt for an AI coding agent.
`MASTER_IMPLEMENTATION_PROMPT.txt`	Identical prompt as plain text.

File	Purpose
`SHOP-DESK-ENGINEERING-BLUEPRINT.docx`	Combined reading edition of the guides and master prompt.
`SHOP-DESK-ENGINEERING-BLUEPRINT.html`	Self-contained, searchable reading edition; no external assets or network calls.
`diagrams/architecture.svg`	Proposed deployment and trust boundaries.
`reference/Legal-Notes-user-supplied.txt`	Your complete attachment, preserved unchanged.
`REFERENCE_SOURCES.md`	Official/technical reading index, with research and applicability qualifications.
`DOCUMENT_VALIDATION.md`	Checks of the documents themselves, not executed SaaS or legal-certification tests.
`PACK-MANIFEST.json`	File hashes for the packaged deliverables; not a software compliance certificate.

How to use the master prompt

1. Give the coding agent the **actual existing repository**, not just screenshots. Supply the user legal notes and this pack.
2. Paste `MASTER_IMPLEMENTATION_PROMPT.md` as its engineering instruction.
3. Ask it to complete **repository inspection, a gap analysis, legal-assumption register and phased plan first**. Review those before authorizing substantial code or financial-data changes.
4. Approve work phase by phase. Require executed tests and evidence at every gate. A design requirement or a green-looking screen is not an implementation.
5. Ask a qualified Indian CA/GST practitioner to validate the tenant-specific tax/document rules and accounting mappings. Obtain licensing/privacy/legal review for the applicable business and contracts.
6. Use separate development/staging databases and approved fixtures. Never test invoices or refunds in production books.
7. Do not declare the system ready for a tenant or transaction type whose necessary legal controls are unsupported. An excluded integration does not remove the tenant's legal obligation.

Existing software versus this target

The reference SQLite release is **1.2.0**. A separate native-MongoDB reference is **1.3.0**. Both are single-clinic references, not verified multi-tenant SaaS products. The inspected MongoDB server code has no `tenantId` implementation and its model list does not include a dedicated audit-event collection. Reusing that code without a tenant/security refactor would be unsafe.

Existing feature behavior is a preservation reference, **not proof that the proposed SaaS has been implemented or tested**. The task completed by this pack is documentation and prompt preparation. New SaaS runtime tests, hosting verification, tenant onboarding and professional approvals remain future work.

Legal and research boundaries

The attachment is now complete; the quantity-only inventory requirement in section 8.3 is included. The pack adds conditional requirements concerning medicine sale/dispensing, MRP/price controls, electronic

records, cash receipts, data protection, cyber incident handling and retention. It does not reconstruct invented missing text or claim to enumerate every Indian or state law.

The legal matrix differentiates an observed official rule, an applicability question, a phased/future requirement and a project-specific safeguard. Enactment, commencement, a Council recommendation and sandbox availability are not interchangeable. All legal rules must be rechecked at the real release date.

Your attached file's SHA-256 is `ade7d2b1abfb979d28b448f36509eaa3bb253a2ff14ff234c8b96ce9c9e27a14`.

02 Product overview and operating guide

shop desk · The Good Vets · Proposed multi-tenant online edition

This guide describes the target product, not a claim that a new SaaS is already running. Configuration, legal applicability and permissions determine which controls can be activated. All examples below are training arithmetic, never production seed data.

1. What the software is

shop desk is a browser-based veterinary counter-management system connecting products, batches, purchases, sales, payments, returns, inventory and double-entry books. Different clinics use the same platform but must never share business records, credentials or financial reports accidentally.

A **tenant** is an independent business workspace. A **branch** is a location within that tenant. A **GST registration** identifies the supplier registration used for a transaction; it is not simply the tenant's display name. One tenant may have several branches and registrations, but a new installation can start with one of each. Transfers between registrations are disabled until their tax and document workflow is approved.

The product is not a veterinary diagnosis tool, prescription-authoring system, payment custodian, tax adviser or automatic GST-filing service. A veterinarian/clinic may still have professional recordkeeping obligations in another approved system.

2. Who sees what

Role	Typical access	Restrictions
Tenant owner	Business setup, staff, approvals, returns/refunds, accounts, reports, exports and recovery requests	Cannot silently rewrite issued invoices, disable the audit trail, bypass law or inspect another tenant.
Manager	Only the explicitly delegated sales/inventory/approval permissions	No automatic owner powers; refund and tax-setting permissions are separate.
Cashier	Own drafts/bills, permitted receipts, shared minimal customer/pet directory, own approved return-note reprints	No stock corrections, tax-classification changes, cost/margin reports, journals, unrestricted exports or refunds by default.
Inventory operator	Products/batches/receipts and quantity-only inventory views if granted	Purchase cost, valuation, selling prices and billing totals require their respective permissions; hiding a column alone is insufficient.
Accountant/read-only reviewer	Authorized ledgers, working papers, period reviews and exports	Posting, reversals and tax eligibility approvals require additional permissions. No staff/password administration by default.
Platform operator	Tenant lifecycle, plan entitlements, deployment health and incident coordination	No default access to clinic invoices or customer data. Exceptional support access must be approved, time-limited and audited.

Names must identify the actual person. Do not share the owner login among cashiers. An owner's membership in one tenant never grants access to another. Staff deactivation must revoke sessions, including already captured sessions used in later requests.

3. Navigation and visual presentation

The tenant and active branch must remain visible at the top of the application. Use a dark, readable system font, clear labels and borders, restrained colors, keyboard focus and accessible errors. The desktop sidebar itself must scroll independently so Accounts, Reports and Settings remain reachable on short screens. Mobile navigation must have the same permitted features.

Suggested navigation:

```
Overview
Billing & POS
Returns & credit/debit notes
Inventory
Purchases & supplier returns
Customers & pets
Suppliers
Payments & cash close
Accounts & ledgers
Reports
Staff & permissions
Settings & compliance readiness
Audit & exports
```

Show only permitted entries, but enforce every restriction again in the API. Consequential actions need an explanation, confirmation and final backend status. Do not show an “issued”, “paid”, “refunded” or “compliant” badge optimistically before the relevant commit/confirmation.

4. Screen-by-screen feature specification

4.1 Overview

Show: selected period, gross billed, customer credits/returns, net billed, confirmed receipts, outstanding balances, refundable amounts, low-stock/expiry alerts and unresolved operational exceptions. Owners may see cost and valuation cards; other users must not receive those values through the API.

Use: choose Today, 7 days, This month, Last month, Financial year, All time or custom From/To. Drill into the contributing source documents. A cashier dashboard is limited to that cashier's authorized work, not the entire clinic's sales.

Manage: every metric needs a definition and a drill-down. Invoice date, receipt date and return date are different measures; do not compare them as if they were the same event.

4.2 Settings and compliance readiness

Show: legal/trade name, address, state/UT, GST status, registration(s), financial-year series, business timezone, receipt methods, invoice templates, permissions, retention rules and readiness issues.

Use: owner enters the actual supplier identity, confirms GST status (regular/composition/unregistered as applicable), obtains classification/document review and selects supported transaction profiles. Configure each series before billing. Store approval evidence and effective dates.

Manage: tax changes create new versions, not changes to old invoice snapshots. Expired licences, unsupported e-invoice obligations, unreviewed tax rules or missing mandatory recipient fields must block the affected sale, not merely display a dismissible warning. “Not required” must be a reviewed applicability conclusion, not a synonym for “not configured”.

4.3 Products and services

Show: name, SKU, barcode, goods/service type, category, UOM, pack conversion, permitted quantity precision, sale price, inclusive/exclusive mode, HSN/SAC, tax treatment/rule, batch requirements, expiry requirements, manufacturer, applicable restrictions and active status. Owner-only fields include purchase cost and margins.

Use: create a product, select verified classifications, set price and UOM, and save. Search by name/SKU/barcode at the counter. A reference lookup can suggest a classification for review; it must not invent a legally approved HSN/rate or silently activate it.

Manage: archive products instead of deleting referenced products. Maintain effective-dated tax/price history. A service has no stock deduction. A product bundle must not receive a guessed “healthcare exempt” or highest-rate mixed-supply treatment; unsupported composite/mixed supplies are blocked pending review.

4.4 Inventory, batches and expiry

Show: physical location, product, manufacturer batch, labelled expiry and precision, available/reserved/non-saleable quantities, supplier/receipt links, status and movement history. Quantity-only views must contain no unauthorized value, cost or margin fields in HTML, JSON, CSV or PDF.

Use: receive stock through Purchases or an approved opening/adjustment workflow. View batches; FEFO chooses eligible first-expiring stock. Near-expiry filters help review 30/60/90-day horizons. A month/year label must retain that original precision and use a reviewed sell-by interpretation.

Manage: block expired, recalled, quarantined or otherwise ineligible batches. Unknown required expiry blocks sale. Physical counting produces an adjustment document with reason and approval; never type over a historical balance. Every movement is append-only, including internal consumption, wastage and damage/expiry write-off with approved accounting/tax-review treatment. Quarantine/disposal is not the same as a sale, customer credit or supplier refund.

4.5 Purchases and goods receipt

Show: supplier, supplier invoice/date/FY, receipt lines, batch/expiry, quantities and bonus units, costs and discounts where permitted, actual supplier GST components, receipt status, payable and payments.

Use: select supplier, enter the genuine supplier bill, record eligible batches and quantities, review costs/tax, then post receipt. The posting adds stock and the accounting liability once. A repeated submission must return the same result, not add stock again.

Manage: duplicate bill checks use the proper supplier/FY scope. Never split a single supplier GST amount into guessed CGST/SGST or mark it automatically claimable ITC. ITC remains pending/ineligible/eligible according to a separately authorized review. A supplier payment reduces the payable, not purchase quantity.

4.6 Supplier returns

Show: original purchase/receipt, original batches, available quantity eligible to send back, reason, dispatch reference, supplier acknowledgment and supplier credit-note status.

Use: select the original receipt, return permitted quantities, approve the inventory movement and record the supplier's actual response. Expired goods returned to a supplier are a documented return/disposal pathway, never a customer sale.

Manage: a buyer's commercial return/debit memo is not automatically the original supplier's statutory GST debit note. Track the genuine supplier credit note or other approved legal treatment separately. Do not

reduce GST/ITC just because goods left the shelf. No placeholder supplier-return button may be presented as functional before this flow is implemented and tested.

4.7 Customers, pets and suppliers

Show: only necessary contact/address/GST details, billing and settlement history, and basic pet identifiers. Customer and supplier balances link to the same party ledgers used by accounting.

Use: create/select the party before billing where required. A walk-in anonymous sale is allowed only if that document/transaction legally permits it. Collect mandatory recipient information based on value, GST status, goods restrictions and other reviewed rules.

Manage: do not use phone number as an unquestionable unique identity; several people/pets may share one. Pet records are billing references, not consultations or prescriptions. Avoid free-text human health information and unnecessary identity documents. Corrections to a directory do not rewrite past invoice snapshots.

4.8 Billing and POS

Show: searchable products, eligible quantity/batch/expiry, sale price, tax treatment, discounts, customer, document type, place-of-supply decision, backend-calculated totals and receipt entry. Clearly distinguish a draft, blocked validation, issuing/unknown outcome and an issued document.

Use: create draft → add products → choose customer/context → obtain a backend quote → resolve warnings → finalize once → print the committed document. Default FEFO can be overridden only to another eligible batch by an authorized person. Discount ceilings and price overrides require permission and audit.

Manage: the backend revalidates stock, rules, price permissions and draft version at finalization. A stale quote requires review. Disabling the button helps usability but database idempotency and unique constraints prevent duplicates. Reprinting never posts another sale or stock deduction.

4.9 Invoices, receipts and printing

Show: appropriate document title, verified supplier identity, number/date, required recipient and delivery data, HSN/SAC, quantities/UQC, discounts, taxable values, legally applicable tax components, place of supply, declarations, final total and authentic statutory integration fields where applicable.

Use: select Corporate, Compact or Classic A4; choose 58 mm or 80 mm thermal/POS output; preview/download/reprint. Configure the real printer, paper width, 100% scaling and browser headers/footers. Test in staging or with a non-posting print preview, not a fake production sale.

Manage: narrow receipts must still contain their required particulars, with continuation pages if needed. A payment QR is not an IRP-signed GST QR. “Computer-generated” is not a blanket replacement for all legally required authentication. Preserve original snapshots, template version and original artifact; show current settlement separately and label its as-of time.

4.10 Payments and cash close

Show: invoice, confirmed receipt allocations, pending/unverified payment attempts, amount, method, actual date/time, external reference, recorder, outstanding and refund entitlement.

Use: record only money actually received as a confirmed manual receipt. If a UPI/card payment is not confirmed, keep it pending and do not mark the bill paid. Record partial payments individually. A receipt may allocate to several invoices only through an explicitly implemented allocation flow whose sum equals the receipt.

Manage: changes use reversal/correction events, not silent edits. Deduplicate requests. Apply reviewed cash-receipt limits and aggregation across the correct legal entity, payer/day, transaction and event. The system cannot see receipts kept in another system, so outside receipts need reconciliation. No card PAN, CVV, PIN or banking passwords are stored.

For cash close, expected cash equals opening cash plus confirmed cash receipts and transfers in, less cash refunds, payouts and transfers out. Count actual cash, record a variance with reason/approval, and post the appropriate accounting voucher. Sales totals are not cash totals.

4.11 Customer returns and credit notes

Show: original issued invoice/line, original batch allocations, quantity already returned, remaining entitlement, saved original money/taxes, goods condition, reason and a default **Do not restock** choice.

Use: owner opens the original bill → Return products → chooses quantities and original batches → records condition/reason → decides disposition → reviews the credit → confirms. A good, eligible return may be explicitly restocked into its original batch only after suitability/resale checks. Cashiers cannot create returns/refunds by default but can reprint approved notes against their own bills.

Manage: partial and repeated returns cumulatively allocate the original amounts exactly; all earlier returns count toward caps, including non-restocked goods. Do not reprice using today's catalogue. A non-restocked return does not inflate sellable stock or automatically reverse inventory/COGS. Keep a separate physical disposition record for goods awaiting quarantine/return/disposal.

A commercial customer credit and an eligible GST liability reduction are separate values/statuses. Preserve original tax amounts as references, but apply a GST adjustment only when its legal conditions and reporting eligibility have been validated. A late/non-qualifying commercial credit can reduce what the customer owes without reducing output GST.

4.12 Refunds, remaining credits and cancellations

Use refunds: approve the refund entitlement → actually return the money outside the software → record date/method/reference/amount → commit once. A refund record does not transfer funds. Pending/failed refunds are not completed cash outflows. Prevent concurrent over-refunds.

Use remaining-invoice credit: after earlier partial returns, credit only the still-uncredited quantities/amounts, including any separately approved service correction. Do not combine an old whole-invoice credit with new partial returns and reverse the same money twice.

Debit notes / undercharge correction: an authorized owner/accountant selects the original issued sale, records the genuine reason, reviews the additional value/tax and issues a separately numbered linked note under the approved legal workflow. A price-only correction changes no stock quantity. Preserve separate source allocations for later credits/returns, including earlier commercial credits; never edit the original amounts or double-credit the adjustment. Additional goods require their own approved fulfillment/document process.

Use cancellation: discard unissued drafts under policy. For an issued invoice, use the authorized legally permitted cancellation/correction process, with reason, timestamps, linked compensating records and any genuine portal acknowledgment. Never delete the issued snapshot or reuse its number. Do not restock goods merely because an invoice status changes; actual custody and suitability matter.

4.13 Accounts and ledgers — explicitly included

Show: customer/supplier, cash/bank/clearing, inventory, sales/services, sales returns/discounts, GST, expense, capital and opening-balance accounts; journals, reversals, trial balance and basic management statements with clearly stated mappings.

Use: post genuine balanced opening vouchers with the accountant; record expenses with supporting references; create balanced journals; review statements and reverse incorrect posted vouchers with linked opposite entries. Period locks control late postings. Automatic sales/purchase/payment/return entries occur in the same transaction as their source action.

Manage: sum of debits equals sum of credits for every posted journal. No invented opening balances or automatic “balancing” revenue. Non-restock returns do not create a false inventory/COGS reversal. Tax-credit eligibility and non-recoverable tax treatment use approved mappings. Basic P&L/balance-sheet outputs are management statements, not audited statutory financial statements. Payroll, depreciation policy, consolidation, income-tax filing and statutory audit remain outside the initial scope unless separately contracted.

4.14 Reports and exports

Preserve these 12 core tabs: **Summary, Day book, Products, Categories, Cashiers, Payments, GST, GSTR-1 working register, GSTR-3B working papers, Stock, Receivables and Stock log**. Add returns, expiry, purchases/payables and audit/reconciliation views where needed; account statements/trial balance remain under Accounts.

The sales register/drill-down also provides invoice/customer/user detail, cancelled sales, discounts, returns and payment-method totals.

Every report supports approved date presets and custom From/To, branch/registration filters, definitions, drill-down and permission-controlled CSV. GSTR labels must visibly say working papers/registers—not official filed returns or ready-to-file portal JSON. Distinguish gross billed, discounts, commercial credits, eligible GST adjustments, confirmed receipts, refunds, net billed, due and margin.

Historical reports use source event dates and saved snapshots. A receipt today against an older invoice belongs in today's receipts but not today's newly billed sales. Large exports run as durable jobs, remain private and expire according to policy. Quantity-only permissions also apply to downloads and hidden fields.

4.15 Staff, audit, backups and tenant administration

Staff: invite individual users, assign branch/role permissions, require owner/platform MFA, review discount/export/refund rights and disable departed users immediately.

Audit: review actor/time/action, source IDs, safe before/after changes, approvals, exports, permission changes and correction links. The audit trail cannot be turned off by an owner or ordinary application administrator.

Backups: inspect actual last-success time, scope, encryption, India storage evidence where required and restore-test results. An export is not a complete disaster-recovery backup. Tenant recovery must not overwrite other clinics or accidentally import someone else's credentials.

Platform administration: manage tenant activation/entitlements without gaining automatic access to clinic records. Record legal terms/privacy/DPA acceptance versions. Subscription collection is initially manual/externally administered unless a separately approved gateway is built; never pretend the clinic payment ledger is the SaaS provider's subscription ledger.

5. First-time setup checklist

1. Verify the owner identity and create a tenant through an authenticated onboarding/invitation flow.
2. Enter the actual business/branch/registration details; review tax status, required records and product/licensing scope.
3. Complete the compliance-readiness gates and accountant-approved document/tax/ledger configuration.
4. Configure unique FY series and coordinate them with any other billing system using the same registration.
5. Create staff accounts and least-privilege permissions; set printing preferences.
6. Create genuine suppliers/products and reviewed classifications. Import only verified opening stock and balances through controlled tools.
7. Test the whole workflow using **separate staging data**, including failed/unknown outcomes, returns and printer widths.
8. Verify backups, restore, India-required storage/log arrangements and incident contacts before first real bill.
9. Activate production. Do not enter sample sales or silently reset a tenant to make it look empty.

6. Daily, weekly and month-end management

Opening: confirm the correct tenant/branch, connection/readiness status, receipt printer, active staff and any urgent recalls/expiry alerts. Record genuine drawer opening cash if used.

During the day: receive stock through documents; bill only eligible batches; verify manual digital receipts; resolve unknown outcomes through the operation-status screen rather than posting again under a new key.

Closing: reconcile cash and UPI/card/bank references, review pending receipts/refunds, compare net billed versus cash collected, investigate failed jobs and confirm backup success.

Weekly: perform selective physical stock checks, review near-expiry/quarantine/disposal, outstanding parties, staff access and reconciliation exceptions.

Month/FY end: accountant reviews tax working papers, ITC eligibility, credit/debit-note eligibility, trial balance, inventory valuation and locked periods; approve rule/series changes for future transactions. Legal retention survives tenant closure.

7. Worked counter example

Illustrative only: taxable value ₹1,000 at 12% produces ₹60 CGST + ₹60 SGST in an applicable intra-state case, total ₹1,120. A confirmed ₹700 receipt leaves ₹420 due. A valid 30% customer credit of ₹336 reduces net billed to ₹784 and due to ₹84. A later credit of the remaining ₹784 leaves net billed zero and ₹700 refundable; a confirmed ₹700 refund then clears that entitlement.

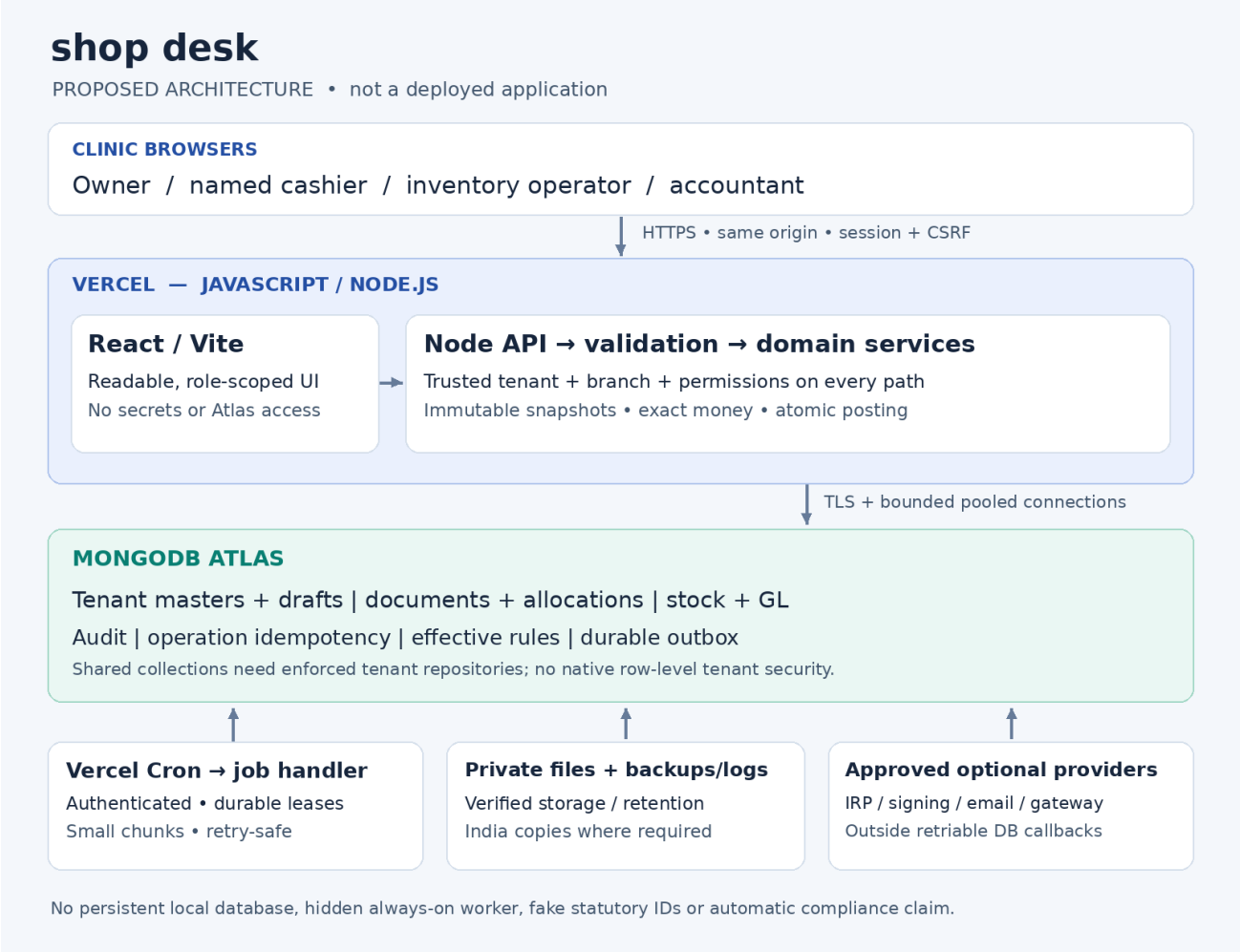
The inventory decision is independent: restock only the explicitly eligible original quantities/batches. GST reduction is independently subject to legal eligibility. The invoice remains preserved throughout.

8. Limits users must understand

- Internet access is required for authoritative billing. No offline sale/payment queue or automatic background financial replay.
- “Record receipt/refund” is a human-entered record, not bank verification or a transfer.

- A tax classification lookup is not professional classification approval.
- An ordinary PDF, hash or payment QR is not an IRN, statutory signature or compliance certificate.
- A feature excluded from this product may still be a legal obligation of the clinic; use an approved external process or block the affected transaction.
- Recovery from an old backup is not permission to reuse document numbers or ignore transactions after the recovery point.

03 Architecture and step-by-step build



Proposed deployment and trust boundaries; actual provider features/location/cost must be verified.

shop desk · Proposed independent-clinic SaaS · Vercel frontend/API + MongoDB Atlas

This is an engineering blueprint. Example interfaces, indexes and command names below are requirements to implement, not a representation that the existing releases already contain them. Preserve the existing archives and real data.

1. Architecture decision record

Decision	Selected design	Why / boundary
Frontend	Existing React/Vite, JavaScript/JSX, ES modules	Preserve working UI/flows; do not switch to Next.js/TypeScript merely for preference.
API	Existing Node/Express service patterns behind Vercel Node.js functions	Same-origin `/api/v1`; no browser connection to Atlas, no Edge-runtime MongoDB assumptions.
Database	MongoDB Atlas replica-set deployment using native Node driver	Real transactions, indexes and validation; no SQLite file in a stateless production function.
Tenancy	Shared collections with mandatory server-derived tenant scope; branches/registrations within tenant	Efficient initial SaaS; requires comprehensive application-enforced isolation because MongoDB has no built-in row-level tenant security here.

Decision	Selected design	Why / boundary
Money	Integer paise with BigInt calculations and BSON Long storage; exact scaled quantities/rational rates	Prevent binary floating-point drift and unsafe JSON conversions.
Financial truth	Immutable documents, allocations, movements, payments and posted journals; mutable derived projections separately	Corrections are linked new records, not overwrites.
Durable work	Transactional Mongo outbox + leased, bounded Vercel Cron invocations	No always-running worker, in-memory-only queue or fire-and-forget finance.
Files/backup	Approved private object storage and provider backups, with actual India-located copies where required	Vercel function disk is not durable. Provider storage/location features must be verified.
Accounting	Full double-entry subledgers/general ledger and trial balance	Explicitly approved extension to legal attachment section 14.
Unsupported legal flows	Fail-closed eligibility gates or a reviewed lawful external workflow	An excluded integration does not remove legal duties.

A database-per-tenant model can be evaluated later for contractual/regulatory needs. It is not a drop-in change: connection budgets, credentials, migrations, restores and costs differ. Do not claim database-enforced tenant isolation for shared collections.

2. Deployment and trust boundaries

```
Owner / cashier / inventory / accountant browsers
      | HTTPS, same origin, secure session + CSRF
      v
Vercel: React/Vite static app + Node.js API
      | request context: user, membership, tenant, branches, permissions
      | Zod validation -> domain services -> tenant repositories
      | Mongo TLS + least-privilege credentials
      v
MongoDB Atlas
  masters | drafts | immutable finance | stock balances | GL
  audit events | idempotency | outbox jobs | legal rule versions
      ^
      | authenticated Cron claims durable leases, small bounded batches
Vercel Cron -> Node job handler -> private artifacts / safe logs / notifications

Separate restricted control plane:
  migrations, access approvals, backups/restores, evidence and incident handling

Optional approved integrations:
  authentic IRP/e-way/signing/payment/email providers
  disabled until actual applicability + implementation + testing are approved
```

The browser never receives database URIs, administrative API credentials, signing private keys, backup credentials or unrestricted customer exports. Platform operations do not automatically gain tenant-data access.

3. Repository reuse and gap analysis

Inspect `/shop-desk` (SQLite 1.2.0) and `/shop-desk-mongodb` (native MongoDB 1.3.0) if supplied. The Mongo reference has native persistence and Vercel routing patterns, but the inspected server has **zero `tenantId`**

occurrences and no dedicated audit-event collection in its model list. It is not a safe multi-tenant starting deployment without refactoring.

Preserve behavior from `src/SidebarMenu.jsx`, billing/inventory/customer/purchase screens, `shared.js`, `returns.js`, `accounts.js`, `reporting.js`, backend domain/accounting/returns/security code, PDF generation and tests. Do not treat numeric SQL-conversion output or old tests as proof of new tenant isolation. Replace all-state bootstrap payloads with permission-scoped resources; never send owner-only costs to cashiers and hide them with CSS.

Suggested incremental layout, not a demand for a needless repository rewrite:

```
src/                        React screens, controls, scoped API client
shared/                     pure DTO helpers, money/quantity functions, formatting
server/
  app.js                    Express routing / Vercel adapter boundary
  auth/                     users, sessions, invitations, CSRF, MFA integration
  tenancy/                  trusted request context, membership and branch guards
  repositories/             only location with normal raw Mongo collection access
  domain/                   billing, stock, purchases, payments, returns, accounting
  tax/                      reviewed policies, quote calculator, validation
  documents/                snapshots, template rendering, signing adapters
  jobs/                     bounded durable outbox consumers
  audit/                    append-only events, redaction, evidence exports
  platform/                 restricted operations, not an owner backdoor
api/handler.js              API entrypoint
api/cron/jobs.js             separately authenticated cron entrypoint
migrations/                 reviewed versioned migrations, NOT public endpoints
tools/                      import, preflight, reconciliation and recovery tools
tests/                      unit, replica-set integration, browser and security
vercel.json                 explicit Node routing, region, durations and cron
.env.example                placeholders only; never real credentials
```

Keep JavaScript/ES modules. Use JSDoc where useful; do not require an unrequested language migration. Pin compatible supported dependencies and maintain a lockfile. Verify supported Node/Vercel/driver versions and runtime end-of-life at launch, starting from the existing Node 22 baseline rather than blindly installing `latest`.

4. Tenant isolation and authentication

1. Authenticate an opaque, high-entropy session cookie. Store only a hash of the session token, with user/membership context and expiry. Use Secure, HttpOnly and an appropriate SameSite policy; enforce CSRF/origin checks for cookie-authenticated mutations.
2. Resolve the active tenant from a server-validated membership/session selection. A tenant slug, route parameter, body field or subdomain is not authorization.
3. Load current membership status/permission version and branch/registration scope. Deactivated staff or changed permissions must lose access without waiting for a stale UI cache.
4. Build a trusted context `{ userId, tenantId, membershipId, allowedBranchIds, permissions, requestId }`. Pass it explicitly through controllers, domain services, repositories, jobs and file access.
5. Every query/update/delete/aggregation/import/export lookup uses the tenant predicate; every related entity must match tenant and appropriate branch/registration. Lookup pipelines must scope both sides.
6. Never spread request JSON directly into Mongo filters or updates. Validate known scalar fields and build permitted predicates server-side. Block operator injection and arbitrary aggregations.
7. Return a non-leaking not-found response for foreign-tenant IDs. Check authorization for the parent and every child allocation, product, pet, batch, payment, journal account and file.

8. Authorize monetary fields at the query/DTO level. Costs, valuation, margins, bank references and full exports have distinct permissions; do not rely on navigation visibility.
9. Keep support access disabled by default. A verified, time-limited, purpose-bound support grant requires approval, narrowly scoped permissions, prominent visibility and audit evidence. No unrestricted impersonation endpoint.
10. Require MFA for owners/platform operators; provide controlled recovery, distributed rate limits, safe password hashing, hashed invitation/reset tokens, session revocation and least privilege.

TTL indexes clean up expired sessions/tokens opportunistically; the API must check their actual expiry. Never add TTL deletion to financial/audit records just to reduce storage.

5. MongoDB logical data model

The following are logical collections. Combine small embedded structures where safe, but do not merge immutable truth into mutable UI state. All tenant-owned documents carry `tenantId`; global control-plane records require separate restricted repositories.

Group / collection	Essential content
`tenants`, `branches`, `tax_registrations`	Actual business identity, branch addresses/state, registration status, owner-approved series, supported legal profiles. Prevent duplicate active GST-registration onboarding across unrelated tenants without controlled review.
`legal_profiles`, `rule_versions`	Applicability, effective dates, references, reviewer/evidence, next review; products reference approved versions.
`users`, `memberships`, `sessions`, `invitations`	Identity separated from tenant membership/role; one person may belong to multiple tenants but permissions do not cross.
`products`, `product_tax_versions`	SKU/barcode/UOM/quantity scale/type, manufacturer/restrictions, price versions, approved HSN/SAC/treatments, active status.
`customers`, `pets`, `suppliers`	Minimal contact/GST/billing reference data; archive history, no clinical chart/prescription authoring.
`batch_lots`, `stock_balances`	Product, branch/location, manufacturer lot, expiry label/precision, received source, conditions, version and exact quantities; cost layers reference genuine receipts.
`sales_drafts`, `purchase_drafts`	Mutable versioned work-in-progress, not statutory invoices or posted stock/GL.
`document_intents`, `number_reservations`	Durable external-integration state and number history when needed; not invented IRNs or finalized invoices.
`financial_documents`	Immutable final invoices/bills/credit/debit notes/purchase records and complete legal/tax/party/line snapshots; `kind` determines legal role.
`document_events`, `document_states`	Append-only cancellation/correction/integration events; mutable rebuildable current status projection, not changes to core financial amounts.
`invoice_allocations`	Immutable original invoice-line/batch quantities and exact cost/tax allocations for returns and COGS.
`returns`, `return_allocations`, `return_dispositions`	Original document/line/allocation links, cumulative amount/quantity controls, commercial and GST-adjustment distinction, explicit restock and physical handling.
`payment_intents`, `payments`, `payment_allocations`	Pending/unverified attempts separate from immutable confirmed receipt/refund records and exact allocations; no gateway custody assumed.

Group / collection	Essential content
`inventory_movements`	Append-only received/sold/returned/transferred/adjusted/written-off quantities, source/actor/time/cost treatment.
`ledger_accounts`, `journal_entries`, `journal_lines`, `period_locks`	Account identity/classification, immutable balanced posted vouchers, source references, exact debits/credits, reviewed openings and period controls.
`purchase_tax_reviews`	Actual input components and authorized eligibility/reclassification evidence; pending is not claimable.
`sequences`, `idempotency_operations`	Atomic numbering scope and durable financial operation identity/result/hash; no premature cleanup allowing replay.
`audit_events`, `retention_holds`	Safe append-only action/correction/access/export/permission history; legal preservation.
`outbox_jobs`, `file_objects`	Durable leased work, private artifact metadata, tenant/actor/purpose/access/expiry, provider idempotency and result references.
`schema_migrations`, `reconciliation_runs`	Controlled version/readiness and checksums/differences; no silent rewrite of source finance.

5.1 Common fields and validators

Use ObjectId IDs internally, UTC instants for events and explicit business dates/timezone (`Asia/Kolkata` initially). Preserve original labelled dates where relevant. Store created/actor/source/schema version, immutable original snapshots, and a version token on mutable projections/drafts. Do not infer the operator from whoever opens the bill later.

MongoDB has no relational foreign-key constraint for these collections. Substitute explicit same-tenant reference checks within transactions, schema validation, unique indexes, controlled writes and archival instead of deletion. These are application-enforced integrity controls, not a claim that native FKs exist.

Set strict `$jsonSchema` validators for required IDs, enumerations and BSON Long amounts; validate nested line shapes and bounded lengths in the application. Prevent raw imports from bypassing validation. Runtime credentials cannot manage validators/indexes or perform unrestricted administrative operations.

Include authorized internal consumption, wastage, damage/expiry write-off and physical count corrections as explicit source-linked movement/voucher types, not hidden balance edits; apply the approved expense/COGS and tax-review treatment. A price-only debit note changes no stock quantity.

5.2 Index requirements

Verify exact query plans and collation before creating indexes. Examples:

```
// Migration examples; collection access and validation omitted for clarity.
await documents.createIndex(
  { tenantId: 1, registrationId: 1, financialYear: 1, numberingScope: 1, number: 1 },
  { unique: true, name: 'uq_document_number' }
);
await sequences.createIndex(
  { tenantId: 1, registrationId: 1, financialYear: 1, series: 1 },
  { unique: true }
);
await operations.createIndex(
  { tenantId: 1, actorId: 1, action: 1, key: 1 },
  { unique: true, name: 'uq_operation_identity' }
);
```

```
await journalEntries.createIndex(
  { tenantId: 1, sourceEventId: 1, postingKind: 1 },
  { unique: true }
);
await stockBalances.createIndex(
  { tenantId: 1, branchId: 1, stockLayerId: 1 },
  { unique: true }
);
await returnAllocations.createIndex(
  { tenantId: 1, returnId: 1, originalAllocationId: 1 },
  { unique: true }
);
await jobs.createIndex({ state: 1, nextAttemptAt: 1, leaseUntil: 1 });
```

Also index tenant/date/document type, customer/due, supplier invoice/FY, movement product/batch/date, ledger account/date, audit tenant/time, stock FEFO eligibility and file ownership. SKU/barcode uniqueness needs normalized fields and partial filters so missing values do not collide. Customer phone is not generally unique. Maintain full-number uniqueness in the legally required registration/FY/document-number scope across its series and coordinate external systems; series alone cannot excuse two invoices with the same printed number.

Registration IDs must map to verified unique supplier registrations. Where a shared legal registration could otherwise be duplicated, enforce a controlled platform-level GSTIN identity/claim process without leaking another tenant's details. If invoices, notes and supplier records share one collection, use fixed server-controlled legal number scopes. Supplier invoice numbers have a separate supplier/FY scope. Do not allow two outward invoices with the same number by adding a freely editable `kind` or scope to an overly permissive index. Preserve genuinely distinct legacy document classes without renumbering their originals.

6. Exact money, quantities and tax

- Keep report aggregation exact too: guard MongoDB Long-overflow promotion to double, use validated Decimal128 intermediate aggregation or bounded BigInt reductions, and return paise strings without unsafe coercion.
- Parse decimal strings; use `BigInt` for calculations. Store paise and scaled quantity units using BSON `Long`, with explicit bounds. JSON transports them as strings. Do not call `Number()`/`parseFloat()` on money or enable driver options that silently promote large values to unsafe JS numbers.
- Define each UOM's scale and legal unit/pack conversion. Prefer explicit integer ratios. A fractional strip/tablet conversion is not authorized dispensing merely because arithmetic supports it.
- Represent rates as exact rational/scaled values with defined precision; do not assume only a small hardcoded list of integer-percentage slabs.
- Quote and finalize through one backend calculator. The browser uses returned amounts; PDF/report generators use saved snapshots, not today's tax functions.
- Define an approved rounding policy for components, lines, invoice discount allocation and final invoice adjustment. Deterministic largest-remainder/cumulative allocation must reconcile to original totals.
- For new applicable intra-state transactions with matching component rates, calculate one half-component under the approved policy and use the equal amount for CGST and SGST/UTGST. Never split an odd combined paise by putting one extra paise into only one component.
- Tax-inclusive fixed prices and equal paise components can produce mathematically incompatible rounding cases. Use only a reviewed explicit taxable/rounding/price policy; otherwise reject the quote and require correction before issue. Never disguise an unexplained difference or alter history to force equality.
- Preserve legacy snapshots, including historical unequal components. Returns/corrections of those documents retain their actual original basis; do not fabricate different historical taxes.

Example boundary helper (real implementation needs validation and tests):

```
import { Long } from 'mongodb';

export function toStoredPaise(value) {
  if (typeof value !== 'string' || !/^-?\d+$/.test(value)) {
    throw new Error('Integer paise string required');
  }
  const n = BigInt(value);
  if (n < -(2n ** 63n) || n > 2n ** 63n - 1n) {
    throw new Error('Money outside storage bounds');
  }
  return Long.fromBigInt(n);
}
```

The UI may use small ordinary numbers for page sizes and dates; the prohibition is against unsafe monetary/quantity arithmetic, not all JavaScript numbers.

7. Atomic billing and idempotency

Every financial mutation requires a random operation key scoped to tenant, actor and action, plus a canonical validated payload hash. Retrying the same operation returns its original result; the same identity with a different payload is rejected. Keep a durable minimal deduplication record for the replay-risk/legal retention period even if optional cached response data expires.

For ordinary supported sales without an external statutory prerequisite:

1. Authenticate/authorize and validate the request size, identifiers and draft version.
2. In a real Mongo transaction, check the idempotency identity and matching payload.
3. Resolve same-tenant products/parties/branches/registration, approved effective rules and legal prerequisites. Revalidate price/discount authority, expiry and stock.
4. Recalculate using the backend calculator and compare the approved quote/draft revision. Do not silently charge a changed amount.
5. Select FEFO eligible stock layers. Use conditional version/available-quantity updates in the transaction. A failed condition causes a clean conflict, not negative stock.
6. Reserve the final number with an atomic counter in its legal scope; no count-plus-one or frontend numbering.
7. Insert immutable invoice/line snapshots, original stock/cost allocations, inventory movements, balanced GL postings, any genuine confirmed receipt, audit event and outbox jobs.
8. Write the durable idempotency result in that transaction. Commit with appropriate majority durability; only then return “issued”.

Pseudocode (not copy-and-run production code):

```
async function finalizeSale(ctx, command, operationIdentity) {
  const payloadHash = canonicalValidatedHash(command);
  const session = mongoClient.startSession();
  let result;
  try {
    await session.withTransaction(async () => {
      // Callback may be retried. No email, PDF upload, portal call or HTTP response here.
      const old = await operations.find(ctx, operationIdentity, session);
      if (old) {
        assertMatchingHash(old, payloadHash);
        result = old.result;
        return;
      }
      const sale = await validateAndQuoteFresh(ctx, command, session);
      const allocations = await decrementEligibleStock(ctx, sale, session);
```

```
      const number = await nextLegalNumber(ctx, sale, session);
      const document = buildImmutableSnapshot(sale, allocations, number);
      await finance.insert(ctx, document, session);
      await movements.insertSale(ctx, document, allocations, session);
      await accounting.postBalancedSale(ctx, document, session);
      await audit.append(ctx, saleIssuedEvent(document), session);
      await outbox.enqueueUnique(ctx, documentArtifactJob(document), session);
      result = committedSaleResult(document);
      await operations.insert(ctx, operationIdentity, payloadHash, result, session);
    }, {
      readConcern: { level: 'snapshot' },
      writeConcern: { w: 'majority' },
      readPreference: 'primary'
    });
    return result;
  } finally {
    await session.endSession();
  }
}
```

All relevant repository calls must actually use the same session. Do not run `Promise.all` inside the transaction. Driver transaction helpers can retry; an unknown commit is not proof of rollback. Bound database/retry budgets to the function's execution budget, handle **only the expected idempotency-index conflict** as a replay, and resolve ambiguous results through a majority-consistent operation lookup. Do not catch every duplicate-key error and pretend success. Never send an Express response from inside a retriable callback. Official driver guidance: 1

7.1 External statutory integrations

Do not put a network call to IRP, signing, gateway, email or object storage inside a retriable Mongo transaction. Use a durable intent with a frozen payload/hash and reserved number, plus an outbox task and provider-specific idempotency/status reconciliation.

When a valid IRN is a prerequisite, the customer's final valid tax document must remain gated until authentic acceptance is recorded. A number/reservation and any stock hold must remain recoverable through unknown provider outcomes; do not release stock or reuse the number while IRP success is still possible. If the external provider succeeded and the next DB commit failed, reconcile the same intent/IRN rather than submit a new invoice. Cancellation/compensation requires the correct actual legal/provider process.

Initial versions can keep these integrations excluded, but then eligibility gates must exclude affected tenants/transactions. A fake “integration successful” adapter is not an acceptable implementation.

8. Returns, credits, refunds and cancellations

Original line/batch allocations must contain the exact sold quantity, discount, taxable value, tax components, gross and cost basis. For an ordinary unadjusted original sale, allocate partial returns cumulatively:

```
new amount = rounded original amount × cumulative returned quantity / original quantity − amount already credited
```

Use a deterministic approved algorithm for every component, with the last eligible return absorbing only legitimate residuals so full cumulative returns exactly equal the original. Do not recalculate from a current price/rate. Later authorized debit/price-correction notes must remain separate immutable source-component entitlement buckets with explicit original-line references. Design and test their allocation, and the effect of earlier non-quantity commercial credits, under the approved policy; the simple unadjusted formula is not a universal algorithm for those cases. Preserve physical quantity caps and avoid crediting any source component twice. A transaction must conditionally consume the remaining return entitlement and, if selected, restore eligible original stock. Concurrent returns cannot both spend the same entitlement.

- Default restock is **false**. Require explicit permission, condition/suitability confirmation and eligible original batch/location before restoring stock.
- No-restock returns do not add saleable quantity or automatically restore inventory/COGS. Record actual physical custody/disposition separately, with zero restored book cost unless an approved explicit valuation/reclassification process applies. Do not later write off that same original cost a second time.
- Commercial credits and output-GST adjustments have separate eligibility/amounts. Preserve original tax reference fields even where permitted GST adjustment is zero.
- A credit note changes the customer's obligation, not the bank account. A refund is a distinct confirmed money record with method/reference/date/actor, permission and idempotency.
- Remaining/full credits consider all previous partial credits and delivered-service constraints. Never reverse an entire invoice twice.
- Cancellation appends legally appropriate events/compensating documents. Draft deletion, issued cancellation, tax credit, goods return and refund are not interchangeable. Keep the original number/snapshot.

Base-case settlement definition, before optional advances/external-credit transfers:

```
net billed = original issued gross + valid debit adjustments − customer credits
net money received = confirmed receipts − confirmed refunds − confirmed receipt reversals
due = max(net billed − net money received, 0)
refund entitlement = max(net money received − net billed, 0)
```

For broader allocation features, model explicit credit applications and their double-entry effects; do not simply subtract a credit twice. Pending payments/refunds have no completed cash effect. Block unexplained overpayments; advances need an approved voucher/time-of-supply flow before activation.

9. Double-entry accounting model

Retain a chart of accounts covering customers/suppliers, cash/bank/clearing, inventory, goods/services sales, sales returns/discounts, COGS, GST input/output, expenses, capital and reviewed opening equity. Derive party subledgers from the same events, not a disconnected manually edited balance.

Examples, subject to approved tax/COA mappings:

Event	Debit	Credit
Goods receipt/invoice	Inventory and eligible/pending/non-recoverable tax according to policy	Supplier payable / genuine immediate payment account
Issued sale	Customer receivable	Revenue and legally applicable output tax
Sold stock	COGS	Inventory at allocated cost
Confirmed customer receipt	Cash/bank/clearing	Customer receivable
Eligible sales return/credit	Sales return and permitted output-tax reversal	Customer receivable/customer-credit account
Commercial-only credit without GST reduction	Approved contra-revenue/expense for the commercial amount	Customer receivable/customer-credit account; **no automatic output-GST debit**
Explicit eligible restock	Inventory at approved original allocated cost	COGS; no such entry merely for a non-restock return

Event	Debit	Credit
Confirmed refund	Customer receivable/refund liability as mapped	Actual cash/bank account
Supplier payment	Supplier payable	Cash/bank
Approved stock loss	Loss/expense and relevant reviewed tax adjustments	Inventory; physical disposal tracked separately
Expense/payment	Approved expense/asset/tax accounts	Cash/bank/payable

Every posted journal balances exactly and has a unique source-event/posting identity. Lines reference same-tenant active accounts; corrections are linked reversal plus replacement where appropriate. Posted entries cannot be edited/deleted. Period locks, controlled reopen approvals, actual date and recorded timestamp remain visible. No fake “balancing” income/expense is inserted to make a trial balance pass.

Manage opening stock and ledger openings as one coordinated controlled migration so inventory is not counted twice. Opening receivable/payable allocations must reconcile to party balances. Basic P&L and balance-sheet working outputs need reviewed mappings and opening/period logic; do not label them audited statutory statements.

10. API contract and frontend state

Representative routes, all requiring backend tenant/permission checks except specifically public authentication endpoints:

API area	Required operations
`/api/v1/session`, `/tenant-switch`	Login/logout/revoke, current scoped membership and allowed navigation; switch only to verified membership.
`/products`, `/customers`, `/pets`, `/suppliers`	Paged/searchable read, validated create/update/archive with field-level permissions.
`/stock`, `/batches`, `/inventory-adjustments`	Eligible availability, batch trace, approved stock events; no raw balance overwrite endpoint.
`/purchases`, `/supplier-returns`	Draft/review/post/return; genuine supplier notes and tax-review links.
`/sales-drafts`, `/sales-drafts/:id/quote`	Versioned drafts and side-effect-free backend quotes.
`/sales-drafts/:id/finalize`	Idempotent atomic issue; no client-specified final tax, number or actor accepted.
`/documents/:id`, `/documents/:id/print`	Immutable saved artifact/snapshot and separate as-of settlement; tenant-owned object authorization.
`/documents/:id/return-preview`, `/returns`	Original remaining entitlement, credit/disposition review and atomic posting.
`/payments`, `/refunds`, `/credit-applications`	Separate pending/confirmed, exact authorized allocations, caps and source references.
`/accounts`, `/journals`, `/period-locks`	Allowed reads, balanced posting/reversal, openings and close/reopen approvals.

API area	Required operations
`/reports/:tab`, `/exports`	Date/branch/registration filters, definitions, restricted DTOs and private durable jobs.
`/operations/:key`	Authorized durable result/status lookup after uncertain request outcomes.
`/audit`, `/settings/legal-profile`, `/staff`	Restricted audit, reviewed rules, invitations/permissions and readiness.

Illustrative finalization:

```
POST /api/v1/sales-drafts/{draftId}/finalize
Idempotency-Key: a-cryptographically-random-operation-key
If-Match: the-reviewed-draft-version
Content-Type: application/json

{"quoteHash":"approved-backend-quote-hash"}
```

Return committed document ID/number, operation reference and source status. A pending external operation can return 202 only after its durable intent exists. Use 401/403, non-leaking 404, 409 for state/stock/quote/idempotency conflicts, 422 for validation and a safe dependency-unavailable/unknown-outcome code as appropriate. Public errors include a request ID, never a Mongo URI/credentials/stack/tenant data.

Use cursor pagination and bounded queries, not an all-record bootstrap. Disable automatic retries of financial mutations under a new key. Preserve a minimal nonsensitive operation key across an uncertain UI outcome; poll/resolve the original authorized operation. Service workers must not queue sales/payments offline. No optimistic finance/stock success. Safe reads/quotes can retry under explicit policy.

11. Reports and printing architecture

Preserve Summary, Day book, Products, Categories, Cashiers, Payments, GST, GSTR-1, GSTR-3B, Stock, Receivables and Stock log. GSTR outputs are clearly labeled working registers/papers until an actually verified filing capability is separately authorized.

Reports use immutable source events plus controlled derived projections with a freshness watermark. Define event dates, inclusive UI date ranges converted to half-open UTC intervals in the business timezone, financial-year boundaries, cancellations, commercial versus tax credits, pending money, cost/restock effects and branch/registration grouping. For an as-of report, later events must not leak backward into earlier balances.

Use the same saved snapshot for A4 and 58/80 mm output. Store template version and authentic integration payloads where required. Render immutable original artifacts; if displaying current settlement, label its as-of timestamp and do not overwrite the original invoice total. Every narrow format retains legally required particulars, even if it spans pages. PDF/email jobs are idempotent artifacts of a committed document, not a second posting path.

12. Durable jobs, files and serverless constraints

Outbox insertion occurs in the same DB transaction as its source action. Jobs have tenant/source/purpose, unique event key, payload version/reference, state, attempts, next attempt time, lease token/expiry and result. Cron claims due work atomically, processes small bounded chunks, and checkpoints progress. Only the current lease token may commit results; a crashed worker's expired lease can be reclaimed safely.

Handlers must be idempotent; assume at-least-once invocation, overlap and partial failure. Use bounded backoff, dead-letter/manual-redrive review and provider-specific status reconciliation. “Exactly once” is

achieved for a financial database event using transactions/unique identities, not promised for every external email. No in-memory job list, local persistent backup, startup timer or unawaited process controls core correctness.

Large exports use consistent defined snapshots/watermarks, pagination and private object storage. Do not put large exports or backup archives through a normal buffered API response. Vercel imposes function/body/execution limits that vary by runtime/plan; verify current limits and keep normal bodies conservatively below them. 1

Private file metadata includes tenant, object key, uploader, purpose, allowed readers, content type/hash, size, retention and scan state. Server generates object keys and scoped upload permissions. Never trust a client-provided tenant prefix or public URL. Validate type/size and scan external documents before use; prevent SSRF. Short-lived download capabilities must be minted only after authorization; avoid PII/secrets in paths and logs. Verify object storage region/versioning/immutability features rather than assume all Vercel products store in India.

13. Security, audit, privacy and retention

Use schema/input validation, safe filtering, output encoding, secure response headers/CSP, strong password hashing, secure session/CSRF controls, MFA, distributed rate limiting and least-privilege infrastructure. Protect login, setup, reset, invitations, exports, support access, refunds and expensive reports. Ordinary setup must not be an unauthenticated global “first caller becomes owner” endpoint.

Financial/audit collections should be read/insert-only for runtime database roles where feasible; mutable drafts/balances/projections use separately scoped privileges. Migrations use separate credentials outside normal API handlers. Retain an independently protected audit/evidence copy and privilege-change monitoring. A hash chain alone cannot prevent an administrator rewriting the database.

Audit actors, events, source IDs, safe before/after fields, approvals, role/config changes, exports, restores and request correlation. Do not log passwords/tokens, raw sensitive request bodies or full payment credentials. Make the accounting audit trail non-disableable. Read/export of sensitive customer/financial information needs authorized access logging with minimized metadata.

Apply the maximum applicable record-class retention/hold, not a global TTL or arbitrary tenant-delete timer. Separate a privacy deletion request from statutory financial retention; restrict/pseudonymize non-required personal fields when lawful. Provide controlled access/correction/export/grievance processes, a data/processor inventory, notices, versioned acceptance and future DPDP readiness. India-required books/backups/logs need actual provider configuration/evidence; `bom1` alone is insufficient.

14. Migration from existing editions

Treat migration as a separately approved operation, never an automatic production startup side effect.

1. Inventory source versions, row counts, monetary storage conventions, invoice series, party balances, batches, originals/credits/returns/payments, users and attachments.
2. Take verified encrypted backups and hash manifests; obtain a rollback/freeze plan and named approval. Do not reset/delete the originals.
3. Create a new target tenant/branch/registration with a migration mapping. Do not attach untrusted source records directly to an existing unrelated tenant.
4. Import to staging with exact historical amounts/rates/date/template/source IDs and actor provenance. Map schema relationships and original batch allocations; preserve cancelled numbers and credit links.

5. Quantize legacy decimal values under an approved conversion/reporting policy. Preserve original stored value representation and an explicit migration reconciliation record; flag anomalies rather than silently “repairing” tax history.
6. Reconcile record counts, full source hashes where applicable, every invoice total/tax, opening/closing stock by batch, customer/supplier balances, receipts/refunds, GL trial balance and inventory valuation. Verify as-of reports before/after.
7. Do not import active sessions, reset tokens, production credentials or another tenant's users blindly. Reinvite/revalidate staff under the new permission model.
8. Reconstruct number high-water marks from issued/cancelled/reserved and external series. Test new numbering without posting a fake real invoice.
9. Rehearse cutoff/delta handling and reversal. If rollback follows new production activity, reconcile those new transactions before any reversal of deployment/data.
10. Run a final controlled migration under freeze, obtain signed reconciliation/approval and only then enable production writes. Keep legacy evidence read-only for retention.

15. Step-by-step implementation sequence

Phase 0 — Inspect, agree scope and preserve evidence

Deliver repository audit, feature-gap table, data-flow diagram, legal-assumption register, risk register and phased plan. Record the full-accounting override. Do not change financial logic/data before the plan is reviewed.

Phase 1 — Foundation and development safety

Set up supported Node, lockfile-based installation, separate development/staging environments, format/lint/unit tests, replica-set integration test support, dependency/license checks, secret scanning and CI. Add empty-tenant/reference initialization only. Define versioned contracts and error types.

Phase 2 — Tenancy, auth and authorization first

Implement trusted context/repositories, memberships, sessions/MFA, invitations, branch/registration scopes, field-level money permissions and platform separation. Build two-tenant adversarial tests before business endpoints are exposed. Refactor old all-state endpoints.

Phase 3 — Schema, exact arithmetic and legal policies

Versioned migrations/indexes/validators; immutable/mutable split; money/quantity/rate helpers; reviewed document/PoS/tax/rounding profiles; quote DTOs; legal readiness gates. Test paise/rounding/overflow/FY/expiry boundaries.

Phase 4 — Masters and inventory/purchases

Products/parties/basic pets, batch receipts and expiry/FEFO, source-linked stock movements, opening/adjustment controls, supplier-return workflow and input-tax review. Add corresponding GL postings and reconciliation, not a later unrelated ledger retrofit.

Phase 5 — Sales and printing

Versioned drafts, backend quotes, atomic/idempotent finalization, registration-wide numbering, original batch/cost snapshots, source-linked accounting, genuine receipt handling and A4/thermal templates. Test concurrency, retries, failure injection and reprint behavior.

Phase 6 — Returns, credits, refunds and correction

Original cumulative entitlement, default no-restock, disposition controls, separate commercial/GST adjustments, cashiers' own-note reprints, remaining credits, separate refund confirmations and lawful cancellations. Test concurrent return/refund caps and no false COGS reversal.

Phase 7 — Accounting, close and reports

Complete chart/openings/expenses/journals/reversals, bank/cash reconciliation, party subledgers, period locks, trial balance/basic management statements and all 12 reports with as-of/event-date semantics. Validate stock/GL/tax/payment reconciliations.

Phase 8 — Durable operations, audit and privacy

Complete append-only audit/privilege evidence, outbox/leases/backoff/dead-letter, private files, exports, retention/holds, privacy/incident workflows, backup verification and restore drills. These controls begin earlier; this phase finishes end-to-end operational coverage.

Phase 9 — Approved integrations or verified exclusions

Implement authentic required integrations only if approved. Test their unknown outcomes/compensation; otherwise prove affected transactions/tenants cannot be activated. Do not hide unimplemented functionality behind enabled buttons.

Phase 10 — Staging, migration rehearsal and production release

Run the acceptance matrix, official Vercel runtime/build/routing tests, two-tenant security review, real printer checks, professional configuration review, India storage/log evidence, backup/restore and migration rehearsal. Publish an honest result/failure list. Deploy only after written technical/business/legal-policy approval; monitor early production and preserve a safe rollback plan.

16. Required developer deliverables

Working JavaScript source, lockfile, migrations/indexes/validators, `.env.example`, tests and raw results, architecture/API/data dictionary, legal applicability register, onboarding/user guide, migration/reconciliation and backup/restore tools, Vercel/Atlas instructions, CI, threat/permission matrix, template fixtures and limitations/release checklist. No secrets, production databases or demo business records in the deliverable.

The complete acceptance and attachment traceability are in `TEST_ACCEPTANCE_AND_TRACEABILITY.md`; online steps and recovery procedures are in `VERCEL_ATLAS_OPERATIONS.md`.

04 India legal and release matrix

Research cut-off: 26 October 2026 · Engineering requirements, not a legal/tax opinion

No prompt, code generator, disclaimer or passing test suite guarantees compliance with every Indian law. Applicability depends on supplier identity, legal form, state, GST status, turnover history, actual supply, product category, medicine licences, data roles and business conduct. A qualified Indian CA/GST practitioner and relevant legal/licensing specialists must verify the actual configuration and releases. Professional review is a project release safeguard; this document does not invent a universal statutory requirement that all software be certified by a CA.

1. How legal rules enter the product

Maintain an effective-dated **legal-assumption register**. Each rule records its official instrument/section, source URL and retrieval date, publication and commencement evidence, territorial/transaction applicability, interpretation owner, reviewer/evidence, valid-from/to, supported cases, required tests and next review. States are `unverified`, `reviewed_applicable`, `reviewed_not_applicable`, `future_not_effective` or `unsupported`.

Do not equate a Council recommendation, enacted amendment, portal advisory, draft rule, sandbox feature or blog explanation with an operative legal requirement. Unknown applicability is not “not required”. Block an affected transaction until its necessary controls or a reviewed lawful external workflow exist. Ordinary tenant owners cannot override statutory gates merely by accepting a warning.

The controls below are **proposed for the build**. They have not been implemented or legally approved for a new SaaS by creating this pack.

2. Core GST/document controls

LEG-01 — Supplier status and correct document

Configure regular GST registration, composition status and unregistered businesses as distinct profiles. Verify the actual GSTIN/legal identity/status and date range; syntactic checksum validity alone is not official registration verification. An unregistered person must not collect GST as if registered. Composition and exempt-supply document logic must be reviewed; do not label every printout a GST tax invoice.

Support tax invoice, bill of supply, appropriate ordinary commercial sale record, credit/debit notes, receipts and cancellation/return records. Where legally applicable, invoice-cum-bill-of-supply is a distinct reviewed capability, not a generic mixed-cart shortcut. Required fields depend on document and transaction facts. Rule 49 specifies bill-of-supply particulars. 1

Release evidence: reviewed document eligibility/field matrix, genuine tenant identity approval, blocked invalid combinations, templates tested on A4 and thermal widths. Preserve electronic-invoice signature exceptions only where actually applicable; do not issue a blanket “no signature ever required” statement.

LEG-02 — Invoice numbering and particulars

Rule 46 includes a consecutive serial number, no more than 16 characters, permitted letters/numbers/hyphen/slash, unique for a financial year, plus transaction-dependent supplier/recipient, description, quantity/UQC, tax, place-of-supply and other particulars. Recipient particulars at specified values and requested particulars below those values must follow the current rule. 2

Build: server-reserved series per legal registration/FY, full-number uniqueness, no reuse after cancellation/reservation, FY rollover tests and a number register. All prefixes/separators count toward length. Coordinate series with external billing channels. A branch ID is not a substitute for correct registration-wide uniqueness.

Release evidence: concurrency tests, legacy/external number migration, cancelled-number history, recipient-field boundary fixtures and required declarations reviewed by the CA.

LEG-03 — HSN/SAC and rates

No product is legally classified merely because a suggested code matches a regex. Notification 78/2020 specifies 4/6-digit HSN requirements by preceding-year turnover, with the stated relief for qualifying supplies to unregistered recipients; portal validations and later instruments also need review. 1

Build: reviewed classifications and effective-dated rate/treatment versions with source/evidence. A reference catalogue is not a catalogue of approved tenant rates. Do not seed a guessed rate for every medicine or freeze an obsolete list of GST slabs. Validate current notifications for the actual product and supply date.

Release evidence: product-by-product approval or approved classification policy; tests for missing/invalid codes, changed rates, exemption versus nil rating and immutable historic snapshots.

LEG-04 — Veterinary healthcare is not the entire pet business

Entry 46 of Notification 12/2017–Central Tax (Rate), under heading 9983, covers services by a veterinary clinic in relation to healthcare of animals or birds. It is not a blanket exemption for all medicines, pet food, retail supplies, grooming, boarding or software subscriptions. The base notification and current amendments/applicability must be checked for the specific supply. 1

Build: independent, approved goods/service treatments; no automatic healthcare exemption for the whole cart. Composite/mixed supplies, ancillary dispensing and bundled packages require professional classification before activation. The SaaS provider's own subscription supply is separate from the clinic's veterinary supplies.

LEG-05 — Place of supply and tax components

The IGST Act's goods rules distinguish movement, bill-to/ship-to and other cases, including section 10(1)(ca) for unregistered recipients and recorded addresses. 1 Circular 209/03/2024 addresses the specified unregistered-recipient delivery/billing-address issue. 1

Build: a reviewed place-of-supply resolver accepting supplier registration, goods/service type, recipient status, delivery/movement and special-case facts. Service rules require separate review; a customer's home state alone is insufficient. Use SGST or UTGST as legally applicable. For supported new ordinary intra-state supplies with equal component rates, preserve the project's equal CGST/SGST-or-UTGST requirement using an approved rounding policy. Never rewrite historical unequal saved tax components.

Gate: exports, SEZ, reverse charge, unusual bundles, distinct-person stock transfers and other unsupported cases remain blocked until scoped and validated.

LEG-06 — Value, discounts and rounding

Section 15 governs value and discount treatment. The CBIC active text accessed in this research includes Finance Act 2026 changes marked **"w.e.f. yet to be notified"**; a coding agent must not apply bracketed future

text merely because it appears on the page. Verify the operative pre-amendment text and any subsequent commencement notification at implementation/release. 1

Build: one backend exact-arithmetic tax service; explicit before-/at-/post-supply discount eligibility, invoice-discount allocation and inclusive-price residual policy. Do not hide rounding differences by altering an issued snapshot. Distinguish paise-level calculation policy from legally prescribed rounding of statutory amounts. Review the actual rounding/time-of-supply treatment; illustrative rates are not product advice.

LEG-07 — Customer credits are not automatically GST reductions

Section 34 provides for credit/debit notes. Its current text specifies the credit-note declaration cut-off of 30 November following the supply FY or furnishing of the relevant annual return, whichever is earlier, plus conditions on reducing output tax liability. The accessed text also shows a proviso effective 1 October 2025 concerning attributable ITC reversal by a registered recipient who availed it, and other incidence-of-tax cases. A further 2026 insertion is marked awaiting commencement. 1

Build: separate original tax reference, customer commercial-credit value, eligible GST adjustment, eligibility review and reporting status. Do not misdescribe a reporting/declaration deadline as a universal prohibition on making any later commercial credit. Record the real annual-return filing date when relevant. Refund recording never substitutes for a required note or proves GST reduction.

Gate: review late notes, post-sale discounts, recipient ITC evidence and supplier-return legal document roles. A buyer's purchase-return memo is not automatically a supplier-issued statutory debit note.

LEG-08 — E-invoice applicability and authentic IRN

Notification 10/2023 reduces the relevant e-invoicing turnover figure to five crore rupees from 1 August 2023; the applicable parent notification, turnover look-back, supplier exemptions and document/recipient coverage must also be assessed. The official listing describes taxpayers **exceeding** ₹5 crore; preserve the operative wording, not a loose threshold label. 1

A separate GSTN advisory applies a 30-day reporting restriction to covered IRN documents for AATO ₹10 crore and above from 1 April 2025, including relevant invoices, credit notes and debit notes. This is not the general applicability threshold. 5

Build: record verified applicability per registration, authentic IRN/signed payload/QR and cancellation acknowledgments only through approved integration or a properly verified external workflow. A normal PDF or UPI QR is not e-invoicing. Do not deliver a legally ineffective “final invoice” where IRN is mandatory but missing. If this integration remains outside initial scope, restrict the product to eligible supported transactions/tenants rather than waive their obligation.

LEG-09 — New/proposed portal features are not automatically operative law

GSTN's October 2026 advisory on recipient-generated RCM self-invoices describes `RCMDOM`/`RCMIMP` as available for sandbox work, with production rollout, applicability and commencement governed by further notifications/communication. 2

Build: track this as a change-watch item, not as an enabled production obligation based only on a recommendation/sandbox feature. More generally, recheck current notifications after Council meetings before deploying tax changes. Do not fabricate self-invoice IRNs.

LEG-10 — E-way bills, dynamic B2C QR and special supplies

Rule 138 contains conditions for e-way-bill information and consignment-value calculation, with exceptions and state-specific implications. Do not implement a single “every bill over ₹50,000 needs an e-way bill” rule. 5

Gate: determine goods movement, transporter/document facts, state rules, exemptions, current validity/cancellation/portal requirements and any separately applicable dynamic B2C QR obligation. These integrations remain excluded unless approved; block affected dispatches or use a reviewed external process. RCM/advances, cess, exports/SEZ and interstate stock transfers are not silently treated as ordinary domestic retail.

LEG-11 — ITC and inventory losses

Input tax eligibility is subject to the relevant conditions, exclusions, evidence and reversals; a purchase invoice or an equal split of a combined tax figure does not establish claimable ITC. 7

Build: pending/eligible/ineligible review; actual supplier tax components; accountant-approved tax/expense/inventory reclassification. Review mixed exempt/taxable activity, losses/write-offs, gifts and other blocked/reversal cases under the applicable provisions. GSTR-2B matching and return filing are not invented by the application. Preserve source facts even when a professional review later changes eligibility.

3. Books, electronic records and cash

LEG-12 — GST stock accounts, audit trail and authentication

Rule 56 covers stock/accounts, document registers, advances and relevant supply records; it requires electronic edit/delete logs and includes authentication of electronic records by digital signature. Its requirements are not satisfied merely by creating a pretty audit screen. 1

Build: append-only financial documents/movements/journals, immutable change events, role-controlled read/export, reproducible record packages and a professionally reviewed record-authentication/signing workflow where applicable. A SHA-256 hash, scanned signature or software checkbox is not a statutory digital signature. Do not confuse an invoice signature exception under rule 46 with the separate rules for electronic books.

LEG-13 — GST retention and recoverability

Section 36 generally requires relevant GST records for 72 months from the due date of the annual return for the relevant year; specified proceedings/investigations can extend retention to one year after final disposal or that period, whichever is later. 3 Rule 57 requires proper electronic backup and production of readable/authenticated records on demand. 1

Build: record-class retention calculation, legal holds, readable exports, restore tests and preserved financial snapshots. Do not implement “delete all tenant data after 30 days” or a blanket “six years from invoice date” policy. Privacy erasure must coexist with required lawful retention.

LEG-14 — Current income-tax books and India backups

The notified Income-tax Rules, 2026 commence on 1 April 2026. Rule 46(8) addresses the specified electronic books/documents remaining accessible in India and daily backups on servers physically in India. Rule 46(9) prescribes seven tax years from the end of the relevant tax year for the specified records, with the further

reopened-assessment provision. Applicability is tied to the persons/books described in the rule, not every database merely because it is Indian. 1

The primary PDF was retrieved and rule 46 read on PDF pages 37–38. Rule 46(6) also refers to additional medical-profession records, including a daily case register in **Form No. 25** and specified stock inventories. Whether and how these apply to the client's veterinary activity requires CA review. Do not casually substitute an old form number or claim a billing-only product fulfills clinical/professional recordkeeping.

Build: actual India backup evidence, daily-success monitoring, the applicable retention clock and external-record workflow where a required professional register is outside scope. Assess transitional periods under the old/new income-tax legislation with the CA.

LEG-15 — Company books and non-disableable accounting audit trail

For company tenants, section 128 includes double-entry/accrual books and at least eight preceding financial years of relevant books/vouchers, subject to applicable extensions. 2 ICAI's revised 2024 implementation guide addresses the company accounting-software audit-trail requirement effective for the relevant FYs from 1 April 2023, including operation/preservation and tampering considerations. 13

The MCA primary Fourth Amendment Rules, 2022 were also retrieved: they amend access to electronic books to “at all times”, replace periodic with daily backup, and add particulars of a person in India where the provider is outside India. Primary URL: https://www.mca.gov.in/bin/ebook/dms/getdocument?doc=MTcyODIyOTI0&docCategory=Notifications&type=open

Build: no disable switch for accounting audit trails; complete transaction/change history and India-accessible/daily backup arrangements where applicable. Application owners and runtime database credentials must not have routine alteration/deletion rights over immutable financial/audit collections. Exceptional privileged maintenance must be controlled and independently evidenced. Hash chains are tamper-evidence, not proof that a privileged database administrator cannot tamper.

Verification limit: the legacy MCA 2021 PDF URL returned an error page during research. The accounting audit-trail conclusion here uses the official professional-body guide; the current amended primary rule/notifications must still be reviewed before company-tenant release.

LEG-16 — Cash receipt restrictions

The official Income-tax Department text of section 186, Income-tax Act 2025, prohibits receipt of ₹2,00,000 or more in the specified non-permitted modes under the alternative person/day, single-transaction and event/occasion tests, subject to stated exemptions. It is not just a per-invoice limit. 1

Build: approved payment-mode and aggregation rules; entity-wide checks across branches, payer identity where required, split-payment detection, recorded exception evidence and conservative controls when outside receipts are unknown. Do not encourage splitting bills, changing customers or recording cash as UPI to avoid a limit. Review other applicable cash/PAN/reporting requirements and the current rules, including prescribed electronic modes, with the CA. A warning cannot make a prohibited receipt lawful.

4. Veterinary product, pricing and disposal controls

LEG-17 — Drug licences, prescriptions and restricted categories

The CDSCO compilation retrieved is Drugs Rules 1945 as amended through G.S.R. 823(E) dated 17 November 2022 (with its stated effective date). It is **not proof that all amendments through October 2026 have been**

consolidated. Rule 65 includes prescription/recordkeeping requirements, animal-owner particulars for veterinary supply, Schedule H1 separate records with three-year preservation, and Schedule X prescription requirements including a two-year retained copy. Different records/categories have different conditions. 1

Build: product restriction category, tenant/location licence profile and validity/review evidence, required external prescription/order references, dispensing/record prerequisites and stock traceability. Do not equate every clinic with a licensed retail pharmacy or assume an own-patient exemption authorizes an open shop. Digital screens do not automatically replace a prescribed physical/bound register or signature. Obtain state licensing/professional advice on exact current requirements and acceptable record formats.

Scope decision: no prescription authoring/clinical records. If a restricted medicine cannot be lawfully sold using a verified external prescription/register process, **block that medicine sale**. Do not lower the legal requirement to fit the UI. NDPS-controlled categories require separate specialist validation and remain disabled by default; medicinal purpose alone is not an unrestricted permission. 1

LEG-18 — MRP and drug price control

The Legal Metrology packaged-commodity rules define retail sale price inclusive of taxes and restrict sales above the applicable retail sale price. Product/category exceptions and current amendments require review; this is not a rule that every clinic service has an MRP. 1

NPPA/DPCO price-control applicability and current formulation-specific notifications must be considered for affected drugs. Do not reuse an old ceiling-price table as current universal veterinary pricing. 1

The official NPPA listing now includes the 30 June 2026 amendment, S.O. 3516(E), affecting matters including specified pack/price determinations, revised-price communication and manufacturer records. A 2016/2023 base compilation alone is therefore not a complete current source. Manufacturer-specific duties must not be imposed indiscriminately on a clinic retailer. 1

Build: applicable batch/pack MRP, unit conversion and tax-inclusive cap checks, approved price-control evidence/effective dates and blocked over-cap totals. Apply the reviewed currently applicable revised cap/price evidence to affected old stock; the originally printed MRP alone may be insufficient after a relevant price revision. Do not add GST over an already tax-inclusive MRP or disguise an overcharge as a product fee. Validate lawful split-pack dispensing separately.

LEG-19 — Expiry, recall, quarantine and disposal

Bio-Medical Waste Management Rules 2016 include veterinary institutions in their scope and address expired/discarded medicines; subsequent amendments/state procedures also need review. 3

Build: block resale of expired/recalled/ineligible stock; record condition, quarantine, authorized supplier return/disposal and supporting references. Financial write-off is not evidence of lawful physical disposal. Capture the actual approved disposal method/vendor/documentation; do not generate fake disposal certificates or instruct staff to discard medicine into ordinary waste. No-restock customer returns must not restore sellable quantity or create duplicate cost reversals.

5. Privacy, security, contracts and evidence

LEG-20 — DPDP commencement and transitional privacy duties

MeitY G.S.R. 843(E) phases commencement of the DPDP Act; G.S.R. 846(E) rule 1 phases the Rules: rules 1, 2, 17–21 at publication, rule 4 one year after publication, and rules 3, 5–16, 22–23 eighteen months after publication. The notifications are dated 13 November 2025 and the Gazette identifiers show publication on

14 November. Use the actual instruments and confirm precise legal commencement dates/corrections with counsel, rather than silently choosing one blog's one-day-different calculation. As of this pack's date, do not say every substantive duty is already in force. 14 2

The December 2025 corrigenda must be read with the Rules. 1 Assess currently applicable IT Act/SPDI obligations during transition rather than treating future DPDP commencement as permission to ignore privacy/security today. 2

Build now: purpose/data inventory, clear notices, appropriate grounds/consent evidence, minimal collection, access/correction/grievance/erasure workflows with lawful retention, children/guardian handling where relevant, vendor/processor agreements, transfer assessment and incident response. The clinic is normally responsible for its customer processing; the platform may be its processor and a separate fiduciary for staff/platform account/subscription purposes. Validate the actual roles rather than rely on labels.

Animal data is not automatically human health data, but the linked owner's contact and billing information is personal data. Do not collect Aadhaar/biometrics/card credentials or a human medical history by default. Do not become or claim to be a statutory Consent Manager merely by implementing a consent screen.

LEG-21 — Logs and breach response

CERT-In directions cover specified entities/incidents, require the stated reportable cyber incidents within six hours of notice/awareness and secure ICT logs for a rolling 180 days within Indian jurisdiction, along with the other applicable direction/clock/contact duties. 1

DPDP Rules 6–8 contain additional security/data/log retention and breach-information provisions once their applicable phase begins. Rule 7 distinguishes notification without delay from its detailed Board report within 72 hours (subject to the stated extension); this is not a substitute for CERT-In's different duty. Rules 6 and 8 include one-year retention provisions requiring precise scope/retention design. 2

Build: named incident contacts/on-call, early reportable-incident triage, reporting clocks and evidence preservation; minimized safe security logs in an approved India-located sink with the applicable retention. Do not log raw secrets/customer documents or wait 72 hours for every incident. Vercel function region alone does not prove every provider log/backup is stored in India; verify vendor features/contracts and block launch if mandatory arrangements cannot be met.

LEG-22 — Contracts and service responsibilities

Obtain Indian legal review of the MSA/service terms, privacy notice/DPA, client responsibilities, supported/excluded use, price/renewal, incident cooperation, backup/RPO/RTO, termination/export/retention, intellectual property, support-access and liability provisions. Evaluate relevant consumer/e-commerce/intermediary/state business obligations on the actual model; do not assume SaaS billing software is automatically a pharmacy marketplace or automatically exempt from every such law.

Preserve acceptance/version/timestamp evidence. Scope exclusions do not excuse defective covered functionality, negligent engineering, non-waivable duties or unsafe security. Do not publish “100% Indian-law compliant”, “GST certified”, “PCI certified” or “audit proof” without the applicable independent evidence. Ordinary business software is not a professional licence.

LEG-23 — Electronic evidence and signatures

Preserve original artifacts, source IDs, actor/time history, rule/template versions, hashes and a controlled export/custody record. The Bharatiya Sakshya Adhiniyam 2023 has electronic-record provisions including section 63; legal advisers must assess the evidence/certificate requirements for an actual proceeding. 6

A hash helps integrity checking but is not automatic court admissibility, a qualified digital signature, an IRP acknowledgment or proof of truth. Never fabricate an official certificate or backdate evidence.

LEG-24 — Additional-scope review triggers

Reassess law before adding online medicine ordering, delivery, clinical/prescription authoring, controlled drugs, payment aggregation/custody, automated gateway refunds, payroll, SMS/WhatsApp marketing, cross-border transfers, exports/SEZ, multi-registration stock transfers, automatic tax filing or wider clinical data. Collect only the relevant approvals; do not ask every tenant for every possible licence or sensitive identity document irrespective of need.

6. Release evidence and who signs off

Evidence	Responsible reviewer	Release effect
Tenant legal identity, GST status, licence scope and exact supported transactions	Owner plus appropriate CA/licensing/legal reviewer	Blocks affected operation/tenant activation if unresolved.
Tax/document/discount/PoS/credit-note rules, effective dates and rounding fixtures	Qualified Indian CA/GST practitioner	Approves the supported policy version, not universal future compliance.
Accounting mappings, openings, ITC review, close/retention/record authentication	Accountant/CA; company auditor/legal adviser where applicable	Blocks accounting release for unsupported obligations.
Tenant isolation, permissions, exact money, retries, reversals and reconciliation	Engineering and independent QA/security review	Executed test evidence required; no assumption-based pass.
India backup/log arrangements, restore tests, incident contacts and RPO/RTO evidence	Operations/security plus legal applicability reviewer	Blocks public production if mandatory controls are unproven.
Privacy/processor/transfer assessment, contracts and termination policy	Indian legal/privacy reviewer and business owner	Final contract/privacy gate.

7. Explicit research limits / open checks

- Sources were read/retrieved for specification design, not an exhaustive all-state legal opinion or certified legal due diligence.
- Current product GST classification/rates, licence conditions, professional registers and physical/digital acceptance must be reviewed per tenant/product; none is guessed here.
- The CDSCO compilation is dated; current amendments and state licensing directions are not fully consolidated in this pack.

- The consolidated Income-tax Act download returned HTTP 403, while the specific official section 186 page and the notified 2026 Rules were accessible. A separate direct eGazette PDF download failed certificate verification; verification was **not disabled**. Do not treat those failed retrievals as successful verification of the entire current Act.
- The referenced SPDI primary-file link did not parse in the current fetch; the current full text and transition remain release-review items.
- The old MCA 2021 PDF path failed. The 2022 amendment primary source and ICAI guide were accessible; current amended primary rules still need release review.
- DPDP precise calendar commencement computation and all relevant corrigenda/later notifications must be confirmed; relative Gazette wording is preserved above.
- Rule 56 electronic-record authentication, record signatures, statutory invoice exceptions, company audit trails and legally acceptable drug registers require a concrete approved solution, not only hashes.
- A production region in Mumbai is not a guarantee that all vendors' global logs, support systems, email or backups remain in India. Verify each relevant flow and contract.
- All new SaaS implementation, runtime/security tests and professional approvals remain pending. This matrix is an acceptance framework, not a completed compliance certificate.

05 Acceptance tests and attachment traceability

shop desk · New multi-tenant SaaS target · 26 October 2026

Status: this is the required test plan. The proposed SaaS has not been implemented or executed by preparing these documents. No row below should be marked passed until the coding team runs it and records the actual environment, command, result and evidence. Existing reference-edition tests do not prove the new tenant/security/legal requirements.

1. Test layers and evidence

- Pure unit/property tests: money, quantities, rates, allocation/rounding, expiry/FY/date bounds, status transitions and ledger rules.
- Native Mongo replica-set integration tests: real sessions, transactions, validators, indexes, conditional concurrency and failpoints where supported. A mock cannot prove atomic rollback or real unique-index behavior.
- API/auth/security tests: direct HTTP requests bypassing the UI, two-tenant adversarial cases, field-level DTOs, files/jobs and lifecycle permissions.
- Browser/accessibility tests: owner/cashier/inventory/accountant roles, scrollable navigation, empty state, no optimistic finance, failed/unknown outcomes and printing.
- Runtime/deployment tests: official Vercel build/runtime/routing, real HTTPS auth/origin settings, database networking/TLS/pools, Cron auth/overlap and provider limits.
- Operational/legal acceptance: verified source/configuration review, actual storage/log location, migration/reconciliation, printer checks, backup restore and incident rehearsal.

For every scenario retain: test ID, requirement IDs, fixture/version, database/runtime, execution command, timestamp, pass/fail/skip, observed values, logs/artifact location and defect/retest. Distinguish unit proof, integration proof, manual review and professional interpretation.

2. Mandatory acceptance matrix

ID	Scenario	Required result / evidence
T01	Empty production onboarding	Zero sample products/parties/pets/stock/invoices/receipts. Only approved system/reference/COA records and verified owner setup. No hardcoded shared default password.
T02	Cross-tenant identifiers	Tenant A cannot list/get/update/finalize/return/refund/export/print or join Tenant B records, including guessed ObjectIds, child allocations, files and jobs. Non-leaking denial; no stock/GL changes.
T03	Membership/branch lifecycle	Captured sessions lose revoked access immediately under policy; branch limits enforced; owner/platform roles do not bypass tenant checks. Last-owner removal requires safe transfer.
T04	Monetary field permissions	Quantity-only inventory JSON/CSV/PDF contains no costs/values/margins; cashier cannot obtain hidden cost fields via bootstrap/exports/error payloads.
T05	Authentication/security	Secure cookies, expiry, CSRF/origin, MFA, reset/invite replay/expiry, distributed rate limits, XSS/operator injection, upload/SSRF and redacted errors/logs tested.
T06	Exact arithmetic and bounds	BigInt/Long/string round trip, negative/overflow/precision rejection, decimal quantity

ID	Scenario	Required result / evidence
		UOM and pack ratios; no unsafe `Number` money path.
T07	Tax/discount fixtures	Inclusive/exclusive, line/invoice discounts, zero/exempt/nil distinctions, mixed rates and odd-paise cases reconcile under approved policy; applicable equal halves remain equal. Unsupported rounding combinations block rather than fabricate.
T08	Historical immutability	Change product/name/price/tax/party/template after issue; old invoice values, original artifact and return basis remain unchanged. Legacy unequal tax remains as stored.
T09	Document eligibility/fields	Regular/composition/unregistered/exempt profiles select valid documents; missing required recipient/HSN/declarations/PoS blocks affected issue. No GST collected under an unregistered profile.
T10	Legal integrations/exclusions	Covered e-invoice/e-way/restricted-medicine cases block without valid prerequisites; never fabricate IRN/QR/licence/register evidence. Reviewed-not-applicable differs from unconfigured.
T11	Invoice sequence concurrency	Parallel checkouts produce unique numbers within the proper registration/FY/legal scope, ≤16 characters/permitted format; FY boundaries and committed/exposed reservations/cancellations do not reuse numbers. An unexposed rolled-back candidate is not an issued number. Coordinate external series.
T12	Stock/FEFO/expiry races	Parallel sales for the last units allow only valid total quantity. FEFO skips expired/quarantined/recalled/ineligible lots; month-expiry/current-time boundary and explicit eligible override tested. No negative stock.
T13	Atomic sale failure injection	Fail after number/stock/doc/movement/journal/audit writes before commit. No partial committed sale, balance, receipt or GL; no misleading UI success. Test actual transaction sessions.
T14	Idempotency/replay	Same key/hash returns same result; same key/different payload fails; concurrent duplicates, double-clicks, retries, webhooks and repeat draft-finalization cannot duplicate stock/documents/payments/journals.
T15	Unknown commit/response loss	Simulate commit-response loss/timeouts. UI reports resolving/unknown, queries original operation, and recovers one committed result; no new-key replay or false rollback claim.
T16	Purchases/duplicate receipts	Original supplier bill/date/FY duplicate checks, batch quantities/costs and source-linked GL post once; failures roll back stock/payable/tax review together.
T17	Supplier return	Original receipt/batch caps, dispatch/acknowledgment, available/non-saleable handling and received supplier note recorded; buyer memo not falsely treated as seller GST debit note.
T18	Partial/repeated returns	Sum of cumulative quantities and each saved original component never exceeds original. Last return exactly consumes legitimate residuals; current catalogue does not alter the result.
T19	Restock versus no-restock	Default false; unauthorized/expired/damaged/cold-chain-unverified returns cannot become saleable. Eligible original restock restores only approved quantity/cost. No-restock makes no false inventory/COGS reversal or later duplicate cost loss.
T20	Return concurrency and full credit	Simultaneous returns cannot consume the same entitlement; remaining/full credit after earlier partial returns cannot credit the invoice twice, including approved service corrections. Linked debit/price adjustments and earlier commercial-only credits use tested per-source entitlement allocations, not a naive raw-original formula.
T21	Commercial versus GST credit	Credit can reduce customer obligation while GST reduction remains zero/pending/ineligible; legal date/annual-return/recipient-ITC evidence is evaluated

ID	Scenario	Required result / evidence
		under approved policy. Reports/GL do not reduce tax blindly.
T22	Payments/partial settlement	Confirmed receipt allocations reconcile exactly; pending UPI/card is not paid. Different dates/actors retained. Receipt retry/reversal cannot double-cash or double-settle an invoice.
T23	Refunds/over-refunds	Credit note alone makes no cash movement. Separate confirmed refund requires permission/reference and cannot exceed entitlement under concurrent attempts. Pending/failed refunds do not reduce cash.
T24	Cash limit aggregation	Approved section 186 rule boundaries, same person/day, single transaction across days and event tests across branches; split/relabel attempts blocked; exemptions require verified evidence.
T25	Cancellation/compensation	Draft discard versus issued correction distinguished; number/core preserved, actual custody/provider eligibility checked, every stock/GL/payment compensation linked and not repeated. Authorized undercharge/debit notes have their own number/GL/source and price-only notes change no stock.
T26	Double-entry postings	Every journal balances; source uniqueness; correct sale/purchase/receipt/refund/return/restock/loss/expense/opening/tax mappings. No artificial balancing accounts.
T27	Openings/journals/period close	Verified party/stock/COA openings reconcile without duplication; invalid accounts/cross-tenant lines rejected; posted vouchers immutable; reversal and controlled reopen retain history.
T28	Full reconciliation	Movement balances = controlled stock projections; inventory carrying value = mapped GL; party allocation balances = subledgers; receipt/refund totals and source postings agree; internal consumption/wastage/loss vouchers reconcile too. Differences surfaced, not silently repaired.
T29	All 12 core report tabs	Correct defaults/presets/custom ranges, branch/registration/cashier scopes, event dates, gross/credits/net/receipts/refunds/due/margin; as-of results do not include later events.
T30	GSTR working papers	Correctly labeled non-filed working registers; commercial credit versus eligible GST adjustment and reviewed input eligibility separated; no fabricated portal JSON or tax-payment status.
T31	A4/58/80 mm printing	All mandatory particulars visible, no clipped totals/HSN/tax, authentic QR only, repeat print adds no financial effect, original template/artifact retained, escaped text/logo inputs safe.
T32	UI/accessibility	Dark readable text, keyboard/focus/errors, independent scrollable sidebar on short screens, usable mobile menu, no unreachable Accounts/Reports/Settings, accurate empty states.
T33	Audit immutability/privacy	Actor/time/source/before-after/approval/role/export/restore events retained and non-disableable; runtime cannot ordinarily alter immutable collections; independent evidence detects privileged tampering. Secrets never logged.
T34	Outbox/Cron/leases	Crash after commit/before artifact, overlapping invocations, expired lease/stale token, retries/dead-letter/redrive and provider duplicate/status scenarios; no lost financial event or duplicate money action.
T35	Private files/exports	Tenant/role ownership, scoped uploads, size/type/scan, signed-link expiry/access logging, large job pagination; no public bucket or cross-tenant path trick.

ID	Scenario	Required result / evidence
T36	Retention/privacy rights	Correct GST/income-tax/company/medicine/security clocks and holds, lawful privacy restriction/erasure, no financial TTL purge; future DPDP changes tracked without pretending already in force.
T37	Backup/restore	Actual successful India-required backups/log evidence; isolated restore/decryption/reconciliation, post-restore number fence, provider-job reconciliation and session revocation; observed RPO/RTO recorded.
T38	Migration/cutover	Original amounts/tax/allocations/cancelled series/actors preserved; counts/balances/hash reports approved; no imported live sessions or unrelated tenant credentials; rehearsal/rollback handles new activity.
T39	Vercel production runtime	Build/routing/direct refresh/auth/origin/env/pool/duration/payload tests pass on intended runtime. No local persistent data or hidden always-on worker. No TLS bypass.
T40	Release and change gates	Professional applicability evidence, signed supported scope/exclusions, source/rule version watch, incident contacts, contracts/privacy acceptance and honest limitations. All blocking defects resolved or affected capability disabled.

3. Property-based invariants

- Each issued financial source has exactly one authorized posting set; sum of journal debits equals credits in exact paise.
- Sold quantity cannot exceed eligible available stock; restored saleable stock/cost cannot exceed explicitly eligible original returned allocations.
- Sum of all return allocations/components cannot exceed the original source, and a complete valid return exactly equals its original amount basis.
- Confirmed refund cannot exceed the customer's actual refundable entitlement; a credit without refund does not change cash.
- Every tenant reference/lookup/result/job/file remains in the authorized tenant; every money-restricted response omits restricted values, not merely their labels.
- A failed or ambiguous response cannot justify posting the same financial operation twice.
- Changing current reference data cannot change a finalized financial snapshot or an earlier as-of report's source events.
- A legally unsupported transaction cannot become issuable because a UI checkbox, owner override or placeholder integration says “success”.

4. Complete attachment traceability

User attachment: `Legal Notes.txt`, 461 lines, SHA-256 `ade7d2b1abfb979d28b448f36509eaa3bb253a2ff14ff234c8b96ce9c9e27a14`. The unchanged copy is in `reference/Legal-Notes-user-supplied.txt`.

Attachment section	Where it is incorporated	Acceptance coverage
1 Role and objective / inspect first	Master prompt §1; Architecture §3, §15 phase 0; README workflow	T38–T40; plan approval before extensive modification.

Attachment section	Where it is incorporated	Acceptance coverage
2 Fundamental principles	Master §2, §5–§7; Architecture §6–§9; Legal LEG-12/13	T08, T13–T15, T25–T28, T33.
3 India tax/invoice requirements, all subsections	Master §6; Architecture §6/7/11; Legal LEG-01–LEG-16	T06–T11, T21, T24, T30–T31, T40.
4 Draft/product/validation/finalize/print sales workflow	User guide §4.8–4.9; Architecture §7/10/11; Master §4/5	T09–T15, T31–T32.
5 Inventory, batches, expiry, stock ledger	User guide §4.3–4.6; Architecture §5/8/9; Legal LEG-17–19	T12, T16–T19, T28.
6 Cancellation, returns, refunds, notes	User guide §4.10–4.12; Architecture §8; Master §5	T18–T25.
7 Payment recording, partial/repeated and refund separation	User guide §4.10/4.12; Architecture §8/9/10; Master §5	T14–T15, T22–T24, T28.
8 Reports, including §8.3 quantity-only visibility	User guide §4.14; Architecture §4/11; Master §4/8	T04, T28–T30, T35.
9 Database, transaction integrity and relationships	Architecture §4–§7; Master §3/5	T02, T06, T11–T15, T26–T28. Mongo equivalents explicitly replace unavailable native relational FKs.
10 Audit trail	Architecture §5/13; Legal LEG-12/15/23; Master §8	T08, T25, T33, T36–T38.
11 Security and access control	Architecture §4/13; Operations §2/5/6/11; Master §8	T02–T05, T33–T36, T39.
12 Error handling and reliability	Architecture §7/10/12; Operations §11/12; Master §5/9	T13–T15, T34, T39.
13 Data retention, backups and recovery	Legal LEG-13–16/20–23; Architecture §13/14; Operations §9–11	T33, T36–T38.
14 Scope boundaries	README approved decisions; Master §0/10; Legal LEG-08–10/17/24	Full GL explicitly retained by user approval. Clinical/gateway/statutory integration exclusions remain unless separately approved; T10/T40.
15 Testing requirements	This matrix; Master §11; Architecture §15 phase gates	T01–T40 plus property invariants; actual evidence required.
16 UI/UX	User guide §3/4; Master §4; Architecture §10/11	T04, T15, T29, T31–T32.
17 Tax configuration/maintenance	Legal rule register and LEG-03/06–09/20; Architecture §6/13; Master §6	T07–T10, T21, T30, T36, T40.
18 Development workflow	README usage; Architecture §15; Master §1/12	T38–T40; inspect/plan, safe incremental build and honest test report.
19 Non-negotiable instructions	All invariant sections, Master §2–§11	T01–T40; no silent history changes/fake compliance/UI-only completion.

Attachment section	Where it is incorporated	Acceptance coverage
20 Final objective	README scope; User guide complete operation; Master §12	Integrated end-to-end release evidence, not just rendered screens.
Suggested delivery workflow and pre-production checklist after §20	Architecture §15/16; Operations §7–§12; Master §12	Final owner/technical/professional-policy approval with source evidence, migration/backup and actual tests.

Additional explicit user requirements retained: **shop desk** name; readable minimal UI; independent scrolling sidebar; no demo business data; A4/thermal/settings templates; applicable equal CGST/SGST with historical preservation; named cashiers; all 12 core reports/date controls/CSV; full accounting; cumulative original-batch returns/default no-restock/separate refunds; multi-tenancy; JavaScript/MongoDB; Vercel frontend+API; approved external workflows or blocks for obligations outside clinical scope.

5. Conflicts resolved explicitly

1. **Attachment section 14 vs full accounting:** the user expressly chose to retain double-entry accounting. This is the approved extension; do not remove it or claim the original attachment already included it by default.
2. **The Good Vets vs software name:** product is **shop desk**; The Good Vets is client/tenant branding. Statutory identity is independently verified.
3. **Relational constraints vs MongoDB:** no false claim of native FKs/RLS. Use validators, tenant-scoped references, unique indexes, controlled repositories, archival and transactions.
4. **All-Vercel vs always-on workers/files:** no local durable filesystem or hidden persistent Node service. Use Atlas state, durable outbox and bounded authenticated Cron, plus explicitly approved managed storage/providers.
5. **Billing-only vs medicine/professional records:** no clinical authoring module, but required sale/dispensing/register controls cannot be waived. Use approved external records or block the affected category/tenant.
6. **Equal new tax components vs preserved history:** apply an approved equal-half policy to the supported new applicable transaction. Do not falsify old tax, force SGST where UTGST/IGST applies or silently “fix” a historical return basis.
7. **Full legal compliance aspiration vs evidence:** requirements and gates are specified; universal certification is not asserted. Unsupported obligations remain release blockers, not accepted disclaimers.

6. Final go/no-go record template

```
Release / source commit / schema version:
Environment / runtime / providers / regions:
Supported tenant and transaction profiles:
Rule/configuration versions and professional review references:
Executed tests and evidence links:
Open defects and affected capabilities (enabled/blocked):
Reconciliation/migration approval:
Backup/restore/location and measured RPO/RTO:
Security/incident/privacy/contract evidence:
Technical approver / business owner / professional-policy reviewers:
Decision and date: GO / CONDITIONAL ONLY FOR STATED SUPPORTED SCOPE / NO-GO
```

A scope-limited release is acceptable only when the unsupported operation is genuinely unavailable and the allowed operation is independently safe/legal. An unchecked checkbox is not a pass.

06 Vercel + Atlas online operations

Target: Vercel React/Vite frontend + Vercel Node API/Cron + MongoDB Atlas

These steps apply to the **future multi-tenant build after implementation/testing**. They are not instructions to deploy the old single-clinic archive and call it SaaS. No account was provisioned or inspected for this pack; no cost was authorized or incurred.

1. What you need before buying or deploying

- The reviewed new source repository, lockfile, migrations, tests and honest release checklist.
- Owner-controlled Vercel and Atlas accounts with MFA and limited collaborator permissions.
- A commercial-use-appropriate Vercel plan, sufficient Atlas compute/storage/connections/transactions and a verified backup/PITR solution. Vercel's Hobby plan is for personal/non-commercial use, not an assumed free commercial SaaS plan. 1
- A domain, an approved email/auth route if used, private file/backup storage and an India-located log/archive arrangement where required.
- Written approval of ongoing infrastructure/provider charges. Verify actual plan features and prices; do not promise unlimited free hosting.
- The actual tenant legal/tax/licensing profile and supported transaction types. If applicable obligations cannot be met with the selected services, stop activation or obtain approval for a suitable managed component; do not hide the gap.

A cost worksheet should separately estimate Vercel seats/runtime/egress/Cron, Atlas compute/storage/backups, stable egress/private networking if required, domain, private storage/versioning, log retention, email/SMS and professional review. Static egress is a separate paid/plan-dependent decision, not something silently enabled by a tutorial.

2. Set up environments and ownership

Create **development**, **staging/preview** and **production** environments with different databases, credentials, storage prefixes/buckets, secrets, auth origins and integration accounts. Production data must not be copied into a public preview. Synthetic fixtures belong only in a clearly isolated test/staging database; production tenants remain empty until genuine records/openings are supplied.

Document who can deploy, manage secrets, run migrations, approve rule changes, respond to incidents and restore backups. Platform operators are not automatically clinic data administrators. Disable public self-service administrative bootstrap. Use authenticated invitations/private operator setup and verified owner activation.

3. Prepare the repository locally

Start from a **working copy** of the chosen reference, not the protected original archive. Ask the agent to complete phase 0 inspection before code changes.

Typical initial commands, when the existing package supports them:

```
node --version
npm ci
npm run build
npm test
```

Use the project's supported Node version, initially the existing Node 22 baseline. Validate Vercel/runtime/driver compatibility and the maintenance horizon before launch. Do not blindly upgrade every dependency or claim a security audit passed without executing it.

The new build must add and document reviewed scripts such as:

```
npm run lint
npm run test:unit
npm run test:integration
npm run test:browser
npm run db:migrate -- --plan
npm run db:migrate -- --apply
npm run db:verify
npm run reconcile -- --read-only
```

These script names are required deliverables, not a claim they already exist in either old ZIP. Read each tool's database/environment confirmation before executing. Destructive reset tools must refuse production and require explicit approval even in staging. Migrations are controlled one-off operations, never an unrestricted API route or automatic per-request action.

4. Create the Atlas project and database

1. Sign in to Atlas using the verified owner account; turn on MFA and invite collaborators with least privilege.
2. Create a project specifically for the environment. Choose a deployment supporting transactions, production capacity and the required backup/recovery features. A low-cost/test tier is not automatically an approved production finance tier.
3. Select an available India region appropriate to latency and legal requirements, for example a supported Mumbai deployment, and verify actual primary/backup location settings. Do not infer all services' storage location from this choice.
4. Configure alerts for connection/resource usage, storage, replication/availability and backup failures. Review the actual backup/PITR features in the selected account/tier. Features change; do not make a blanket assumption that every small tier has no backup or that every paid tier automatically has every recovery feature.
5. Create a **runtime database user** with only the needed database/collection actions. Prefer insert/read privileges for immutable financial/audit collections and controlled update privileges only for mutable drafts/balances/projections. No cluster-admin/validator/index-management powers in the API.
6. Create separate short-lived/restricted migration/restore credentials. Keep them out of normal Vercel runtime environments and frontend build variables.
7. Obtain the official driver connection string. Keep credentials private and URI-encode password characters correctly. Store the database name separately if the application expects it. Never paste secrets into support screenshots or this document.
8. Configure Network Access for the **actual backend egress**, not merely your laptop IP or the public application/domain IP.

5. Atlas network access and TLS

Atlas IP access lists control which clients may connect. A public website's ingress address is not its backend's outbound address. Vercel serverless egress can vary unless the selected service provides approved stable egress/private connectivity. Choose an actual supported solution and verify the egress IPs for the functions' selected region/environment before narrowing the Atlas list.

Adding `0.0.0.0/0` allows connections from anywhere and increases exposure; it is not the default security recommendation. 1 Any temporary diagnostic exposure needs explicit approval, strong least-privilege credentials, monitoring, expiry and a rollback plan. Do not leave a wildcard allowlist as a silent production shortcut.

On an initial connection failure, diagnose separately: DNS/SRV, network access list/egress, paused cluster/service health, TLS trust/handshake, driver/runtime compatibility, authentication and missing environment variables. A message such as `MONGODB_TLS_FAILED` classifies a failure, not its proven cause.

Never "fix" TLS by setting `NODE_TLS_REJECT_UNAUTHORIZED=0`, `tlsAllowInvalidCertificates`, invalid-hostname acceptance or a plaintext connection. Capture sanitized error category/request ID for operators. Keep raw Mongo URI/credentials/host topology out of public errors.

Use a module-level cached Mongo client/pool compatible with Vercel's lifecycle. Set conservative max pool size, idle limits and selection/operation timeouts based on measured concurrency. Estimate total connections as warm instances × per-instance pool plus jobs/operations; one pool of 10 per instance can still exhaust a small cluster when the app scales. Clear a rejected cached connection promise so one failed startup does not poison the instance indefinitely. Reuse the pool; do not connect/close per request or establish a separate pool for each tenant by accident.

6. Configure Vercel

1. Create/import the reviewed new project from the owner's repository. Do not make a repository public if it contains private code/configuration.
2. Select the correct root directory and Vite/static build configuration, normally install `npm ci`, build `npm run build`, output `dist`. Confirm the actual project scripts instead of guessing.
3. Use the supported **Node.js runtime** for the Mongo API. Do not deploy it as Edge runtime. Set the function region deliberately, normally `bom1` if available and approved, rather than assume the default is India.
4. Preserve/test API rewrites and the SPA fallback. `/api/v1/...` must reach the Node API, not return `index.html`; frontend routes must load after refresh. Authentication/setup/status errors must remain meaningful and sanitized.
5. Give functions conservative tested durations and workload limits. Do not rely on a plan's maximum duration as a job queue. Limits/features change and some are beta. 1
6. Configure a protected production-only cron schedule consistent with the selected plan. Cron execution can overlap or retry; its handler must use durable leases. Authenticate using a strong `CRON_SECRET`/supported authorization mechanism, reject unauthorized requests and do not accept an arbitrary browser tenant ID as the job scope. 1
7. Protect preview deployments and route preview secrets only to staging. Confirm that Cron, log drains and external providers use the correct environment.
8. Add security headers/caching rules. Financial/auth/API responses are private/no-store; static fingerprinted frontend assets can be cached. Avoid putting PII/auth tokens in URLs or public assets.

Environment variable checklist

Final names are a versioned application contract. Typical server-only variables:

Variable	Purpose / rule
`MONGODB_URI`	Secret native-driver connection URI; server-only.

Variable	Purpose / rule
`MONGODB_DB`	Explicit environment-specific database.
`APP_ORIGIN`	Exact approved HTTPS origin for auth/CSRF/link generation.
Session/token hashing secret(s)	High-entropy independent secrets according to the auth design; never static examples.
MFA encryption/key reference	Protect MFA secrets; do not log or expose them.
`CRON_SECRET`	Authenticate scheduled job invocations; not a tenant API key.
Private storage endpoint/region/bucket/credentials	Minimum required permissions; prove location/retention where legally relevant.
Safe log sink/credentials	Approved retention/location; redact secrets and personal data.
Optional approved provider credentials	Environment-specific IRP/signing/email/payment integrations only after approval.

Never prefix secrets with `VITE_`: Vite exposes such variables to the browser bundle. Do not include actual `.env` files in Git/ZIPs. Vercel environment changes usually require the appropriate redeployment; verify the active deployment uses the intended environment, not a stale preview. Use secret-management rotation and incident procedures, not credentials in screenshots.

7. Database preflight and migrations

Before the first live deployment:

1. Confirm the target environment/database fingerprint, available transactions, runtime role, network route, backup and current source schema version.
2. Run a dry-run migration plan against staging with a verified backup.
3. Apply reviewed migrations using separate operator credentials. Create validators, tenant-prefixed indexes, immutable-role privileges and schema/version records. Do not silently rebuild/drop live indexes on every function invocation.
4. Initialize reference metadata/approved chart of accounts only. No products, invoices, customers, pets, suppliers, stock or cash movements are fabricated.
5. Verify indexes, permissions, empty-tenant counts and the read/write capability matrix. A `readWrite` user on the entire database does not prove append-only protection.
6. Rehearse the upgrade/downgrade compatibility plan. Refuse startup/mutations against unsupported schema versions with safe operator diagnostics.
7. Repeat the approved production procedure during a controlled release window, preserving existing data and evidence.

8. Staging verification and go-live

Run the full acceptance matrix with two independently seeded test tenants. Verify backend isolation even when a request deliberately sends foreign IDs; test every API, report, CSV/PDF, file, job and support path. Execute concurrency, idempotency, injected failure, unknown commit, repeated returns, over-refund, historical tax and GL/stock/payment reconciliation tests.

Use the official Vercel build/runtime path as well as local tests. Check login/session/CSRF behavior on the real HTTPS host, direct SPA navigation and API routes, Cron authentication, function budgets, pool reuse, payload limits and protected previews. Verify A4/58/80 mm output with the real printers. Do not create fake production invoices to test printing.

Obtain the CA/GST/licensing/privacy/contract reviews specified by the legal matrix; unresolved mandatory controls block the affected tenant/capability. Test actual India-required backup/log/archive locations, not just UI checkboxes.

Then: final approved migration → owner activation → genuine opening data → production write enablement → close monitoring. Give owners the user guide and contacts. Display support/exclusion limits honestly. Plan expiry/subscription enforcement separately from the clinic's financial ledger, with contractually appropriate secure record access/export/retention after termination.

9. Routine operations

Monitor: API error/latency, failed authentication/rate limits, tenant-scope violations, unknown outcomes, duplicate conflicts, stock/GL reconciliation, queue backlog/oldest job, dead letters, lease recovery, storage/cost growth, backup age, restore evidence and expiring licences/rule reviews.

Daily: confirm provider backups and India daily copies where required; inspect failed jobs/pending provider outcomes; reconcile cash/digital receipts; review critical expiry/recalls and access alerts.

Weekly: check staff/support grants, dependency/security advisories and working backup retention. Run read-only reconciliation of stock balances to movements, journals to sources and customer/supplier balances to allocation records.

Periodic: test full restore and tenant recovery, rotate credentials under policy, rehearse incidents, review vendor/subprocessor arrangements and upcoming legal/rate changes. Roll out future rule versions by effective date without recalculating old invoices.

10. Backup and recovery design

Proposed targets such as **RPO 15 minutes / RTO 4 hours** are engineering objectives, not a promise until the actual paid tier, backup configuration and restore drills demonstrate them. Document provider outage and missing-backup scenarios honestly.

Use verified provider database backups/PITR plus independent protected evidence/archives as required. Cover immutable documents, allocations, movements, payments, GL, audit, rules, numbering history, file metadata and file objects, and the separately protected keys needed for recovery. A CSV export is not a complete database backup. A backup without tested decryption/restore is not proven recoverability.

For mandatory daily India backups/log retention, retain provider location evidence and last-success alerts. Do not attempt an unlimited full-database `mongodump` inside a normal Vercel request. Use the approved Atlas/native managed backup capabilities; bounded artifact jobs are not automatically substitutes. Large one-off import/restore tools may run on an explicitly approved operator machine/CI runner with temporary controlled access, not a hidden always-on product backend. Obtain approval before adding another managed compute service.

Recovery procedure

1. Declare incident, record time/scope, preserve evidence and stop affected financial writes. Prefer read-only mode over destructive guesses.

2. Restore the selected backup/PITR into an isolated restricted environment, never directly over the shared production database without approval.
3. Verify manifests/decryption/schema, stock by batch, original amounts/tax, payment/refund balances, trial balance, source links, rules and number history.
4. Reconcile transactions after the restore point against independent numbering/audit evidence, customer-issued documents, bank/provider records and IRP where applicable. **Do not reuse any number merely because the restored counter is lower.** Restore a safe issuance fence/high-water mark before enabling new bills.
5. Reconcile outbox/provider status. Do not blindly resend payments, refunds, e-invoice submissions or notification jobs restored from an earlier state. Use the original operation/provider identities.
6. Revoke/rotate sessions or compromised credentials according to the incident, including stale sessions that might reappear from backup. Do not import source-tenant credentials into another tenant.
7. For one-tenant recovery, restore/extract through reviewed tenant-scoped tools; do not overwrite unrelated clinics' collections. Preserve approvals and recovery audit evidence.
8. Verify the recovered tenant in read-only mode, obtain business/technical approval, enable writes gradually and monitor. Record actual RPO/RTO and gaps.

A code rollback is not automatically a database rollback. New production invoices/payments after deployment must not disappear because a previous application version is redeployed.

11. Incident response and safe diagnostics

Maintain an actual on-call/contact matrix, reporting applicability and legal clocks. Triage CERT-In-listed incidents and privacy breaches separately; do not wait for the wrong regime's deadline. Preserve security evidence in the approved location, minimize exposure, rotate compromised secrets and record required notifications/decisions with legal review.

A public health endpoint should reveal only minimal generic service readiness. Deep diagnostics require an authorized operator and must never dump customer records, keys, full environment values, database URIs or raw stack traces. Treat symptoms as symptoms; no unverified cause is declared fixed without an actual successful environment test.

12. Stop / no-go conditions

Stop production activation if tenant isolation, exact money, numbering, stock/payment/GL consistency, required legal document/medicine controls, authentication/audit, recoverability or mandatory India arrangements fail. Do not compensate by disabling TLS, relaxing permissions, deleting audit rows, fabricating statutory IDs or rewriting old bills. Publish remaining limitations and obtain the right approvals instead.

07 Copy-ready master implementation prompt

0. Approved product, scope and precedence

Act as a senior full-stack JavaScript engineer, MongoDB/transaction architect, security engineer and QA lead experienced in Indian billing/inventory systems. Build a maintainable commercial product, not only UI screens.

The software name is **shop desk**. **The Good Vets** is a client/tenant brand, not an assumed legal supplier name. Statutory documents must use the actual verified supplier identity.

Approved target:

- Independent-clinic **multi-tenant SaaS**, with branches and legal/GST-registration scopes inside a tenant.
- **JavaScript/ES modules, React/Vite frontend, Node.js API on Vercel, MongoDB Atlas using the native Node driver**. Prefer the existing project structure and compatible libraries. Do not introduce an unapproved framework/language/database rewrite.
- **All application frontend and API hosting on Vercel.** No hidden always-on Node backend/worker, persistent local SQLite database or local durable filesystem. Use Mongo durable state, bounded authenticated Vercel Cron consumers and explicitly approved private storage/providers.
- Veterinary **billing/inventory only**, with customers, basic pet identifiers and suppliers. No diagnosis/treatment/clinical chart or prescription authoring.
- **Full double-entry accounting is INCLUDED by explicit user approval**, overriding only the general-ledger exclusion in section 14 of the attached Legal Notes. Include party/cash/bank/inventory/sales/purchases/GST/expense/opening/journal accounts, reversals, trial balance and basic reviewed management statements. Other exclusions remain.
- Preserve minimal dark/readable UI, a genuinely scrollable left navigation, named cashiers, complete billing/inventory/HSN/batch/expiry workflows, A4/corporate/settings templates, 58/80 mm thermal output, all 12 existing core report tabs/date controls/CSV, cumulative original-invoice/line/batch returns, default no-restock and separate refunds.
- New production tenants start **empty**. Do not create demo/sample products, parties, pets, stock, sales, payments or balances. Approved system references and an account chart are not demo business transactions.

The complete user attachment is `Legal Notes.txt` (461 lines; SHA-256 `ade7d2b1abfb979d28b448f36509eaa3bb253a2ff14ff234c8b96ce9c9e27a14`). Incorporate every non-conflicting requirement, including its section 8.3 quantity-only reporting/permission requirement and its final workflow/preproduction checklist. If a requirement conflicts with the approved scope, explain it and seek approval rather than silently changing it. This prompt consolidates the target; use the attached overview/architecture/legal/acceptance/runbook documents for deeper detail if provided.

Do not claim universal Indian-law compliance, government/GST/PCI certification, professional licensure or an audit opinion. Source verification and professional review must be explicit. Scope exclusions do not excuse defects, unsafe security or non-waivable obligations.

1. First action: inspect and plan, do not rewrite blindly

Before extensive changes:

1. Inspect repository, package/lockfile, runtime, frontend routes/components, backend APIs/auth, database models/validators/indexes, finance/tax/returns, PDF/report generation, tests and deployment configuration.
2. Identify what is genuinely implemented, incomplete, unsafe or contradicted by this specification. Preserve working domain/UI behavior; prefer incremental refactoring with evidence over a needless rewrite.
3. Produce an architecture/data-flow diagram, feature-gap matrix, permission matrix, legal-assumption register, migration/risk plan and phased test-driven implementation plan.
4. Identify existing business data and protected files. Never reset/delete/overwrite production or change historic financial money without an expressly approved controlled process.
5. Explain any missing dependencies/credentials/legal facts; do not invent them. Ask targeted questions only for genuinely unresolved decisions. The scope choices in section 0 are already settled.
6. Wait for plan approval before substantial financial schema/logic changes or external provisioning. Never purchase a plan, enable paid networking or deploy to a live tenant without explicit authorization.

If the provided references are SQLite 1.2.0 and native MongoDB 1.3.0, treat them as **single-clinic reference editions**. The inspected Mongo server has no implemented `tenantId` scoping and its listed models lack a dedicated audit-event collection. Do not rename it SaaS and deploy it unchanged. Old test results do not prove new isolation, audit or idempotency.

2. Non-negotiable integrity invariants

- Financial truth is immutable: preserve original numbers/dates, supplier/recipient/line snapshots, HSN/SAC, UOM, price/discount/tax, totals, batch/cost allocations, actor and rule/template versions. Correct through linked events/documents, never silent overwrite/delete.
- Atomicity: document, stock, allocations, journal, actual confirmed receipt, audit event and durable outbox/idempotency result commit together where they form one business action. Failures leave no partial committed action.
- No overselling/negative stock, expired/ineligible stock sale, duplicate invoice/payment/refund/return/journal, over-credit, over-return or over-refund.
- One authoritative backend exact-arithmetic tax/calculation service; frontend quotes, PDFs and reports cannot independently invent totals. Current catalogue changes never recalculate issued history.
- BigInt calculations, BSON Long paise/quantity units, exact rational/scaled rates and JSON strings. Initial currency is INR; no unreviewed foreign-exchange/multi-currency flow. No binary floating-point money or unsafe `Number`/`parseFloat` coercion. Guard database aggregation/Long overflow and any automatic double promotion; use validated exact Decimal128 intermediates or bounded BigInt reduction for reports, and checked string results.
- Same financial operation retried with the same identity/hash returns the original result; a changed payload under that identity is rejected. Button disabling alone is not duplicate prevention.
- Unknown commit/provider/response outcome is **unknown**, not automatically failed/rolled back. Resolve the original operation before retrying; never replay under a fresh key just because the UI timed out.
- Every posted journal balances; inventory, payments, party balances and reports reconcile to source events. Never insert invented balancing revenue/expenses or silently repair historical truth.
- No demo business data, hardcoded owner/password, fabricated statutory identifiers, fake gateway acknowledgments, placeholder functional buttons or unverified “tests passed” claims.

3. Application boundaries, data model and tenancy

Keep business logic in tested domain services, not React components or request handlers. Controllers authenticate/validate/authorize; repositories enforce trusted tenant scope; services run transactions; pure helpers calculate exact amounts; documents/reports consume immutable sources.

3.1 Trusted context and authorization

- Derive `{userId, tenantId, membershipId, allowedBranchIds, permissions, requestId}` from authenticated server-validated membership. Never trust a body/header/subdomain tenant ID as authorization.
- Pass context explicitly. Never keep a mutable global “current tenant” or reuse tenant-specific state across concurrent Vercel requests. A shared Mongo client/pool is fine; a shared mutable request context is not.
- All collection queries, updates, aggregations, lookup joins, child/reference checks, imports, exports, jobs, signed file links, caches and support tools must enforce tenant and branch/registration scope.
- Mongo has no native FK/RLS for this design. Use strict validators, tenant-prefixed unique indexes, same-tenant references checked within transactions, controlled repositories and archive-not-delete behavior. Do not claim SQL FKs exist.
- Reject unknown fields/operators; do not spread request JSON into queries/updates. No client-supplied arbitrary aggregation, raw query, `$where`, file path or remote fetch URL.
- Authorize monetary fields before DTO serialization. Quantity-only inventory views/export/PDF must not contain unauthorized cost/valuation/margin. Navigation/CSS hiding is not security.
- Enforce current staff status and permission version; revoke captured sessions after disabling staff. Secure tenant switching only for actual memberships. Protect last-owner transfer/recovery.
- Platform operators have no default clinic-data access. Exceptional support grants require verified purpose, owner/authorized approval, expiry, narrow permissions and audit; no invisible universal impersonation.

3.2 Logical collections

Design and document these concerns; embed bounded related structures only where appropriate:

- Tenant/branch/registration identity and entitlements; legal profiles and effective-dated rule approvals.
- Global users plus tenant memberships, secure sessions, invitations/reset/MFA metadata.
- Products/tax versions/UOM and price policies; minimal customers/basic pets/suppliers.
- Batch/receipt cost layers, stock balances/reservations and append-only inventory movements.
- Mutable versioned sales/purchase drafts; durable external document intents and number reservations where required.
- Immutable final financial documents, original line/batch/cost allocations; append-only document events and separately rebuildable status/settlement projections.
- Original-linked returns/allocations/disposition; commercial credit versus eligible GST-adjustment amounts/status.
- Pending/unverified payment attempts separate from immutable confirmed receipts/refunds and exact allocation records.
- Ledger accounts, immutable posted journal entries/lines, period locks and input-tax-review evidence.
- Sequences, durable financial idempotency identities/results, append-only audit, retention holds, leased outbox jobs, private file metadata, migrations and reconciliation evidence.

Use ObjectId references internally, UTC event instants, explicit business dates/timezone and preserved expiry label/precision. Snapshots include actor IDs and original display values without leaking costs in ordinary sale APIs. Avoid unbounded arrays and oversized transactions/documents.

3.3 Indexes and least privilege

Create reviewed migrations with required-field validators and tested indexes: legally correct document-number uniqueness by tenant/registration/FY/approved number scope; sequence scope; finalization/source-draft uniqueness; normalized SKU/barcode partial uniqueness; supplier invoice/FY duplicate checks; stock layer/branch uniqueness; source-event journal uniqueness; idempotency identity; FEFO/search/report/ledger/audit/date access and due-job leasing. Use explicit permitted number scopes, not a client-editable kind that permits two tax invoices with the same number.

Coordinate registration identity and numbering across branches and external billing channels. A duplicate GSTIN/registration claim across unrelated tenants needs controlled verification without leaking another tenant. A single legal receiving entity's cash-limit aggregation cannot be narrowed merely by creating branches or workspaces.

Runtime credentials have minimal read/insert privileges for immutable financial/audit collections and controlled updates for mutable state where supported. No routine delete/update privileges over source finance; no validator/index/admin access. Use separate restricted migration/restore credentials outside normal APIs. TTL cleanup is for expiring auth/ephemeral records, never a shortcut to deleting books/idempotency evidence prematurely.

4. Screens, roles and everyday usability

Build a minimal readable React interface: dark high-contrast text, simple system fonts, labeled inputs, clear totals/errors, keyboard/focus support, compact responsive layouts and independent sidebar scrolling. All features must remain reachable on a short screen and mobile. Show tenant/branch context persistently; do not rely on color alone for state. Never display optimistic financial success.

Required modules and management behavior:

1. **Overview:** permitted gross billed/credits/net/confirmed receipts/due/refund entitlement, low-stock/expiry/exceptions with period definitions/drill-down. Cashier sees authorized own work, not owner totals.
2. **Billing/POS:** product/SKU/barcode search, backend availability/quote, customer/pet reference where needed, eligible batch/expiry/HSN/tax, permitted discounts, clear draft/blocked/issuing/issued/unknown states.
3. **Documents/printing:** correct legal title/fields, Corporate/Classic/Compact A4 and 58/80 mm thermal templates, immutable originals and no stock/finance effects on reprint. Settings version templates; no arbitrary unsafe scriptable template editor.
4. **Inventory:** reviewed products/classifications/UOM, manufacturer batches/expiry/locations/stock status, FEFO, movements, approved counts/adjustments, quarantine/recall/disposal; quantity-only permissions end-to-end.
5. **Purchases:** original supplier bill/date/FY, actual supplier tax components, received lots/expiry/quantity/cost, payable/payment and separate ITC review. Duplicate posting cannot add stock twice.

6. **Supplier returns:** original receipt/batch eligibility, dispatch/reason, supplier acknowledgment/actual credit note and approved legal/accounting treatment. Do not label a buyer memo as an automatically valid seller statutory debit note.
7. **Customers/pets/suppliers:** minimum necessary directory/contact/GST/address fields and source-linked statements; basic pet billing identifiers only, no clinical chart/prescription authoring. Archive referenced masters.
8. **Customer returns/notes:** original invoice/line/batch remaining entitlement, condition/reason, cumulative components, default no-restock, permissioned suitable restock and separate GST eligibility.
9. **Payments/refunds/cash close:** actual confirmed receipts and partial allocations, pending/unverified attempts, separate refund recording, method/reference/date/actor, cashier drawer reconciliation and variance approvals.
10. **Accounts:** party/cash/bank/clearing/inventory/revenue/purchases/tax/expenses/capital/openings, balanced journals/reversals, trial balance, reviewed basic P&L/balance-sheet working outputs, period locks and reconciliation.
11. **Reports:** preserve exactly these 12 core tabs: Summary, Day book, Products, Categories, Cashiers, Payments, GST, GSTR-1, GSTR-3B, Stock, Receivables, Stock log. Add returns, expiry, payables and audit views as needed without dropping the originals.
12. **Settings/staff/compliance/audit/exports:** verified identity/GST profile/series, rule versions, readiness blockers, invoice preferences, individual invitations/permissions/MFA, non-disableable audit and private exports/retention/recovery status.

Roles: owner, explicitly delegated manager, named cashier, inventory operator, accountant/reviewer and separate platform operator. Default cashier restrictions exclude returns/refunds, stock adjustments, classifications, costs/margins, journals and unrestricted exports; allow reprinting approved notes against their authorized own bills. Any broader delegation needs a named permission, not an accidental owner-style payload.

First-use onboarding must be authenticated, owner-verified, empty and gated by necessary configuration. No unauthenticated global “first caller becomes owner” setup. Provide practical first-day, daily close, weekly expiry/access and month/FY-end instructions. SaaS subscription entitlement/admin is separate from clinic customer money; no automatic gateway subscription collection is implied.

5. Financial workflows and transaction implementation

5.1 Quote and finalize

Drafts are versioned, not issued invoices. A quote is computed on the backend from same-tenant references and applicable rule versions. Revalidate all facts at finalization: stock/expiry/status, price/discount authority, supply/recipient/PoS, mandatory fields, licensing/e-invoice prerequisites and draft/quote version. Do not silently accept a changed total; return a reviewable conflict.

For a supported ordinary sale, one Mongo transaction must:

1. Resolve the operation identity and matching canonical payload hash.
2. Validate same-tenant references and legal readiness; recalculate exact approved totals.
3. Allocate eligible FEFO stock and perform conditional available-quantity/version updates. No count-then-write race, negative stock or expired-batch override.
4. Atomically obtain the legal registration/FY series number; no frontend or count-plus-one generation.
5. Insert immutable complete financial/line snapshots and original batch/cost allocations.

6. Insert movements, balanced source-unique GL entries, any actual confirmed receipt, audit and unique outbox events.
7. Record the durable idempotency result; commit majority-durable and only then respond issued.

Every repository operation uses the same session. Do not use `Promise.all` or network/PDF/email/storage side effects inside a retriable callback. Do not send an HTTP response inside it. Bound transaction/driver retries to the function budget. Handle the specifically expected idempotency conflict, not every duplicate-key error, as a possible replay. Driver guidance: 1

Keep finalization/source-draft uniqueness so a new key cannot finalize an already issued draft twice. Operation status lookup remains authorized. If a response is lost, preserve the original key and resolve it; no new financial queue/retry on reconnect.

5.2 Numbering and state

Reserve/generate server-side; enforce full printed-number format/length, approved scopes and FY behavior. Track committed/exposed issued/cancelled/reserved/abandoned numbers and never silently reuse them. Coordinate with other systems using the same legal registration. An unexposed candidate rolled back with its whole transaction is not an issued or durably reserved number. Separate immutable document truth from mutable state projections; business lifecycle, statutory acceptance/cancellation, settlement and physical fulfillment are distinct dimensions.

5.3 Purchases, inventory and adjustments

Posting a genuine receipt creates stock/cost layers, movements/payable/tax-review/accounting once. Direct product quantity editing is not an opening/receipt/adjustment workflow. Document actual supplier tax; never split a single total into invented input components or auto-claim ITC. Partial/bonus receipts and supplier returns require exact quantity/cost policy and source links. If a partial receipt/accrual flow is not implemented, block it instead of inventing a full receipt.

Month/year expiry retains its precision and reviewed sell-by cutoff; recalled/quarantined/cold-chain-unverified/damaged stock is ineligible even if a calendar expiry is future. Required unknown expiry blocks sale. Backdating a new counter sale must not bypass current custody, expiry or period controls. Disposal records actual approved handling, not merely a UI balance subtraction. Support authorized internal consumption, wastage and damage/expiry write-offs as explicit movement/voucher types with approved expense/COGS/tax-review treatment, not hidden quantity edits.

5.4 Returns and credit entitlement

Use the original finalized invoice, line and sold batch allocations. All previous partial, full, commercial and no-restock credits consume the correct entitlement. For the ordinary unadjusted sale, cumulative monetary allocation must reconcile exactly:

```
new component = rounded(original component × cumulative returned quantity / original quantity) − previously allocated component
```

Use an approved deterministic implementation with proper caps/residuals. Current product prices/rates cannot affect old return amounts. The last valid return must reconcile original quantities, discounts, taxable value, components and gross—not create an extra credit from rounding.

Authorized later debit/price-correction notes are separate immutable source-component entitlement buckets linked to the original line. Explicitly design/test their return allocation and the effect of earlier non-quantity commercial credits; the simple unadjusted formula is not universally sufficient. Never edit originals,

over-return physical quantity or credit a source component twice. Provide a reviewed undercharge/debit-note workflow with correct numbering/tax/GL and no stock movement for a price-only change.

Restock defaults false. Only explicitly approved suitable goods in an eligible original batch/location can increase saleable stock and reverse the appropriate original allocated cost/COGS. A non-restocked return creates no false inventory/COGS reversal; record physical returned/quarantine/disposal custody separately and avoid a second cost write-off. Refuse ordinary product-return credits for an already delivered service unless the separate authorized correction workflow permits it.

Commercial customer credit and eligible GST adjustment are separate. Keep original tax reference versus actual GST adjustment/status/evidence; late or ineligible commercial credits can change customer balance without lowering output GST. Remaining/full credit considers earlier returns and cannot reverse an invoice twice. Concurrent returns conditionally consume remaining entitlement in the same transaction.

5.5 Payments, refunds and cancellation

Confirmed manual receipts mean staff actually verified money received; pending/unverified UPI/card is not paid. Record method/reference/date/amount/actor and exact allocations. No card PAN/CVV/PIN/bank password. Duplicate reference checks follow the actual provider/account semantics; split allocations do not become duplicate receipts.

A credit does not send money. A refund is a separate permissioned, idempotent confirmed outflow after actual external payment, capped transactionally by real entitlement. Pending/failed refunds do not reduce cash. Support receipt reversals/corrections with history; block unexplained overpayments/advances until an approved receipt-voucher/time-of-supply flow exists.

Base supported-case settlement: net billed = original gross + valid debit adjustments − customer credits; net received = confirmed receipts − confirmed refunds − confirmed receipt reversals; due/refund entitlement are positive differences in the appropriate direction. Explicit credit transfers/advances require separate allocation/accounting design, never double subtraction.

Draft discard is not issued cancellation. Issued correction/cancellation needs permission/reason, preserved number/core, lawful eligibility, genuine external acknowledgment if required, linked financial/stock events and separate refund. Cancellation cannot magically return goods to custody or erase bank movements.

6. India tax, medicine and legal-policy requirements

Verify current official instruments at implementation/release and obtain tenant-specific review. Maintain a versioned legal-assumption register with source URL, publication/commencement, applicability, interpretation/reviewer/evidence, effective range, supported/blocked cases and review date. Store rule/template versions on originals. Clearly differentiate `unverified`, `applicable`, `not applicable after review`, `future not effective` and `unsupported`.

6.1 Tax/document engine

- Distinguish regular GST, composition and unregistered profiles. Do not collect GST without entitlement or assume every document is a tax invoice. Implement legally relevant tax invoice, bill of supply, reviewed ordinary commercial record, notes and receipts/vouchers.
- Validate supplier/recipient/description/HSN-SAC/UOM/discount/value/component/PoS/declaration fields by actual transaction; not every field is universally mandatory. Rule 46 numbers are no more than 16 characters with the specified permitted characters and uniqueness. 2

- Do not invent HSN/SAC/rates or treat classification regex/reference suggestions as professional approval. Verify turnover/recipient-dependent HSN detail and current notifications/portal rules. Keep exemption, nil rate, zero rating and out-of-scope categories distinct.
- Entry 46 of Notification 12/2017–Central Tax (Rate) concerns veterinary-clinic animal/bird healthcare, not all medicines/retail/grooming/boarding/software. Verify exact current classification and composite/mixed supplies. 1
- Determine goods/services place of supply from actual applicable rules and facts, not merely customer home state. Review bill-to/ship-to, unregistered delivery-address and service cases separately. Apply CGST/SGST, CGST/UTGST or IGST lawfully. Unsupported special cases block.
- For supported **new** applicable equal-rate intra-state cases, CGST and SGST/UTGST paise amounts must be equal under an approved rounding policy. Never silently assign an odd paise to only one half. If fixed inclusive price/component equality cannot reconcile lawfully, use a reviewed explicit policy or reject before issue. Preserve legacy amounts and original correction bases, including unequal saved components.
- Validate discounts/time of supply and rounding. Do not recalculate originals after rule updates. The CBIC section 15/34 pages accessed for this specification show 2026 amendments marked awaiting notification; recheck actual commencement rather than activate displayed future text automatically. 1

6.2 Notes, integrations and special transactions

- Apply current section 34 conditions and reporting cut-off: 30 November after the relevant FY or relevant annual-return filing, whichever earlier, and applicable tax-incidence/recipient ITC-reversal conditions. Distinguish issuing a commercial credit from entitlement to reduce output GST; preserve evidence/status and operative effective dates. 1
- Verify e-invoice supplier/document/recipient applicability, parent notifications, look-back and exemptions. Notification 10/2023 uses the relevant **exceeding ₹5 crore** context; do not confuse it with the separate **AATO ₹10 crore and above / 30-day IRP reporting** restriction from 1 April 2025. 1 5
- Never invent an IRN, signed QR, cancellation acknowledgment or e-way bill. Ordinary PDF/UPI QR ≠ authentic e-invoice. Covered final document delivery remains blocked until its necessary statutory prerequisites are valid.
- External calls belong outside retriable DB callbacks: durable frozen intent/number/hash → outbox → provider idempotency/status lookup → controlled completion. Unknown provider success must not free reservations/reuse numbers or produce a second invoice. If external success precedes a DB outage, reconcile the same identity; compensate only through the actual lawful process.
- E-way movement rules, dynamic B2C QR, RCM/advances, cess, exports/SEZ and inter-registration transfers require their own current applicability/design. Do not generalize one ₹50,000 rule or ordinary-retail tax flow to all cases. Keep unsupported affected operations disabled.
- October 2026 recipient-generated RCM self-invoice sandbox features (`RCMDOM`/`RCMIMP`) are a change-watch item pending the relevant production/legal commencement; recommendations/sandbox availability are not automatically operative law. 2
- Input-tax eligibility is separately reviewed; actual supplier tax components, evidence, exempt/taxable common inputs, blocked credit/reversals and GSTR-2B review matter. Purchase tax is not automatically claimable ITC.

6.3 Medicine sale, MRP and external records

- Verify tenant/location/product-specific drug sale/dispensing licences, personnel, prescription/order and register requirements under current central/state law. Do not assume an own-patient clinic exemption authorizes open retail.
- Record necessary external references and genuine sale/dispensing particulars even though clinical/prescription authoring is excluded. Review legally acceptable physical/digital register/signature formats. The retrieved CDSCO rule 65 compilation includes animal-owner particulars, H1 separate records/three-year preservation and X prescription/retained-copy conditions; it is an older compilation, not a complete 2026 amendment check. 1
- H/H1/X/NDPS or other restricted medicines remain blocked unless their actual licence/external prescription/register/controlled-stock process is implemented and reviewed. No fake licence/clinical approval checkbox.
- Enforce applicable tax-inclusive MRP and verified current NPPA/DPCO price controls/pack conversions. Do not add GST above an inclusive applicable MRP or guess a current ceiling. Review the 2026 DPCO amendment and actual revised-price/batch evidence, not only an old base compilation or originally printed MRP; do not impose manufacturer-only duties on every retailer. Verify lawful split-pack dispensing.
- Prevent expired/recalled/damaged/ineligible resale, quarantine unsafe returns and record approved supplier-return/disposal evidence. A write-off is not lawful physical disposal; never generate fake certificates.
- Where income-tax/professional rules require a case register or other records outside this product, document an approved external process or block unsupported onboarding. Do not claim billing screens replace clinical/professional records.

6.4 Cash, retention, signatures and privacy commencement

- Apply reviewed Income-tax Act 2025 section 186 restrictions: the official text addresses receipt of ₹2,00,000 or more in the specified non-permitted modes using alternative person/day, single-transaction and event tests, subject to stated exemptions. Aggregate by actual receiving legal entity across relevant branches and payments, not just per invoice/customer-row. Detect split/relabel attempts; require verified payer/exception evidence as needed and reconcile outside-system receipts. Review other current PAN/cash/reporting provisions. 1
- GST records generally use section 36's 72 months from relevant annual-return due date, with proceedings/investigation extensions/holds. Do not substitute invoice-date six years. Rule 56/57 stock/accounts/edit-log/authentication/backup requirements need actual controls. A hash/scanned signature is not a legally recognized digital signature; review rule 56 electronic-record authentication separately from rule 46 invoice exceptions.
- Notified Income-tax Rules 2026 rule 46 includes specified electronic-books accessibility in India, daily India-server backups, seven-tax-year retention and reopened-assessment provisions; additional medical-profession records (including Form 25) need professional applicability review. Do not rely only on superseded rule numbers/forms. 1
- Company tenants need current company-books/audit-trail/India backup/access/retention review, including non-disableable accounting edit history and applicable eight-FY records. An application owner or runtime credential must not routinely alter immutable history. Independent evidence/privileged-change controls are necessary; hash chains are not an audit certification.

- As of the specification research date (26 October 2026), DPDP Act/Rules 2025 have phased commencement. Rules 1/2/17–21, rule 4 after one year and rules 3/5–16/22–23 after eighteen months follow the official publication/commencement instruments and corrigenda. Confirm exact dates and later changes; do not say all substantive duties already apply or ignore current IT Act/SPDI duties. 2 1
- Design notices/purposes/appropriate consent or other permitted grounds, rights/grievance/access/correction/erasure with lawful retention, children/guardian handling, processor/vendor contracts, transfer assessment and incident evidence. Do not collect Aadhaar/biometrics/human medical history/card credentials by default. Do not claim to be a statutory Consent Manager from a consent UI.
- Assess CERT-In applicability: listed reportable incidents within six hours and covered ICT log retention for 180 days within Indian jurisdiction, plus required contact/time-sync duties. Future-applicable DPDP security/processing retention and breach duties are separate, including their one-year provisions and prompt/detailed-report distinctions. Do not wait 72 hours for every incident. 1
- Keep maximum applicable record-class retention and legal holds; no automatic whole-tenant purge on subscription expiry. An India function region alone does not establish India storage of every backup/log/vendor flow.
- Preserve electronic evidence/artifact hashes/custody/source and actual signatures as applicable. Hash ≠ court admissibility certificate, official IRP evidence or professional legal opinion.

7. Complete double-entry accounting extension

Implement a genuine transaction-linked ledger, not financial-looking cards or a disconnected manual spreadsheet. Maintain customer/supplier subledgers and cash/bank/clearing/inventory/sales/services/returns/discounts/COGS/GST/expense/capital/opening accounts with reviewed mappings.

- Purchase/receipt: inventory and appropriate eligible/pending/non-recoverable tax treatment versus payable/actual payment; never automatic input-credit eligibility.
- Sale: receivable versus revenue/output tax; allocated COGS versus inventory.
- Receipt/supplier payment: actual money account versus the party balance.
- Eligible customer credit: approved contra-revenue and only eligible tax adjustment versus party credit; commercial-only credit must not automatically debit output GST.
- Explicit eligible restock: restore original approved cost/inventory versus COGS; no such reversal merely for no-restock returns.
- Refund/loss/expense/opening/transfer/journal events: balanced source-linked mappings and actual evidence; no duplicate cost loss after a no-restock return.
- Every posted journal uses same-tenant valid accounts, exact balanced lines and unique source-event/posting identity. Immutable after posting; reverse/correct with links. No fake balancing entry.
- Openings require genuine reviewed inputs and coordinated party/stock balances; no double-counting inventory. Period locks and controlled reopen preserve both actual business date and recorded time.
- Reconcile trial balance, stock carrying value, party allocations, cash/bank and tax working balances. Basic management P&L/balance sheet need reviewed account mappings and must not be labeled audited statutory statements.

Payroll, depreciation policy, consolidation, income-tax filing and statutory audit are not silently included just because the GL is included. Document them as future scope unless separately approved.

8. Reports, audit, privacy and security

Every report supports Today, 7 days, This month, Last month, Financial year, All time and custom From/To, applicable branch/registration/cashier filters, metric definitions, drill-down and permission-controlled CSV. Preserve event-date/as-of semantics: issue date, receipt date, return date and refund date differ. Date queries use reviewed business-timezone half-open UTC bounds.

Provide invoice/customer/user sales detail and cancelled/return/discount/payment-method drill-downs. Separate gross billing, discounts, commercial credits, eligible GST adjustments, net billed, confirmed receipts, refunds, due, refundable amount and actual cost/margin. Non-restock returns do not falsely restore profit. Clearly label GSTR-1/3B outputs working registers/papers, not filed returns/ready portal JSON. Historical stock and balances reconstruct from events or verified snapshots with freshness/source watermarks; later events do not leak into an earlier as-of report.

Provide non-disableable append-only audit for finance/stock/corrections, rule/permission changes, sensitive access/exports, support grants, migration and restore. Record actor/time/source/request/reason and safe before/after values; never passwords/tokens/card credentials/raw sensitive bodies. Record failed privileged attempts in a safe operational/security trail without fabricating a committed financial posting. Protect independent evidence against privileged tampering and document residual administrator risk honestly.

Use maintained authentication patterns, strong password hashing, secure HttpOnly/Secure cookies, appropriate SameSite/CSRF/origin checks, owner/platform MFA, controlled invitations/reset/recovery, secure headers/CSP, output escaping, distributed rate limits and restricted operational endpoints. Validate request sizes, file type/size/scan and SSRF-safe uploads; no arbitrary URLs. Do not make costs/private customer data available in bootstrap payloads, error messages or downloadable hidden fields.

Files/exports are private, server-keyed and tenant/role/purpose-bound. Authorize before minting short-lived downloads; record access, enforce retention and scan untrusted attachments. Preserve original document/template/artifact versions; reconstruction is not falsely presented as an identical original file. Privacy rights and legal retention operate together. Publish reviewed service/privacy/DPA/support/termination terms and acceptance versions; no disclaimer replaces security or professional/legal responsibilities.

9. Vercel, Atlas, durable jobs and operations

- Browser calls same-origin relative APIs; database/keys never enter `VITE_` variables or the frontend bundle. In preview/deployment, never point browser code at localhost for its backend.
- Use Node.js functions, correct API/SPA routing and an explicitly selected approved region; test direct frontend reload and API paths in the actual Vercel runtime.
- Reuse a bounded Mongo pool across warm invocations; avoid per-request connect/close or a pool per tenant, clear failed cached connections, and size total connections for real function concurrency. Pin supported runtime/driver/dependency versions and verify lockfile/build/audit.
- Verify actual Atlas backend egress/allowlist/private-network options. Never disable TLS verification. A generic TLS failure is not proof of a single cause. Broad `0.0.0.0/0` access is not a silent default; stable egress/private networking has plan/cost implications requiring approval.
- Vercel functions are bounded/stateless. No durable local disk, in-memory-only queues, startup timers, unawaited financial effects or hidden persistent worker. Verify current function/body/response/plan/Cron limits. Commercial SaaS is not assumed eligible for a free personal-use plan.
- Insert outbox jobs transactionally with the source event. Authenticated Cron atomically claims small due batches using lease tokens/expiry; checkpoint bounded work, reject stale lease commits and tolerate

overlap/crash/retry. Backoff/dead-letter/redrive require controlled review. Downstream handlers/provider calls use source identities/status reconciliation; no unqualified external “exactly once” promise.

- Large exports/private PDF artifacts use bounded jobs and approved private storage; no huge synchronous buffered response or full-database backup in a normal function. Temporary file use is not durable storage.
- Use separate dev/staging/prod secrets/databases/storage/provider accounts; previews cannot use live tenant data. Record cost/location/subprocessor features and approval before enabling services.
- Monitor latency/errors, scope violations, unknown outcomes, pool pressure, reconciliation differences, queue age/dead letters, backup age/restore evidence and expiring legal rules/licences.
- Verify real backups/PITR, India-required daily copies and security-log retention. Exports are not complete backups. Document measured RPO/RTO; proposed targets are not guaranteed without drills/tier evidence.
- Restore into isolation, reconcile money/stock/GL/rules/files and transactions after the restore point, restore a safe document-number fence before new issue, reconcile external jobs/providers and revoke stale sessions as needed. Do not overwrite other tenants, reissue old numbers or resend financial actions blindly.
- Migrations/backfills/imports/restores require dry-run/backup/target confirmation, counts/hashes/balance reports, rollback/cutover approval and separate restricted credentials; never mutate all production data on startup. Preserve source archives and historical snapshots. Reinvite staff rather than import live sessions/secrets.
- Operational diagnostics are sanitized and authorized; deep status never dumps environment secrets, connection strings, customer records or raw stacks. Provide actual incident contacts/reporting clocks/evidence retention.

10. Initial exclusions and honest capability gates

Keep excluded unless separately approved: clinical records/prescription authoring, online pharmacy marketplace/delivery, controlled-drug workflows without validated prerequisites, payment aggregation/custody/automatic gateway refunds, automatic GST filing, payroll/consolidation and unimplemented statutory integrations/special supplies. An approved external workflow must be concrete and lawful; a generic checkbox or disclaimer is insufficient.

If an excluded capability is legally mandatory for a tenant/transaction, disable that transaction or restrict onboarding until a verified solution exists. Do not tell the user that an excluded integration removes legal obligations. Do not broaden scope, charge for services or claim an external account was configured without authorization.

11. Required executed tests and release evidence

Use the existing test framework where practical, plus real replica-set Mongo integration, direct API/security, browser/accessibility, official Vercel runtime and operational/manual tests. Mocks alone cannot prove transactions/indexes. Maintain a requirement-to-test matrix and record command/environment/time/result/artifact. Do not mark unrun tests passed.

At minimum execute:

1. Empty tenants/no demo; authenticated onboarding; safe owner recovery.

2. Two-tenant adversarial access across every read/mutation/child/lookup/report/PDF/CSV/file/job/cache/support path; current permissions/branch revocation; restricted money absent from payloads.
3. Money/quantity/rate bounds, overflow, odd paise, inclusive/exclusive, discounts, equal applicable halves, immutable old rates/amounts and full residual return allocation.
4. Document/GST profile/HSN/PoS/required-field boundaries; unsupported statutory/medicine workflows blocked; genuine integration failure/duplicate/unknown outcome where implemented.
5. Parallel invoice numbering/FY/series, last-unit stock FEFO/expiry/quarantine races, duplicate receipt/finalization, same/different idempotency payload, injected transaction failures and lost commit responses.
6. Purchases, supplier returns, cumulative partial/no-restock/eligible-restock/remaining credits, concurrent return caps, separate GST eligibility, partial/pending receipts, refund caps/concurrency and lawful cancellation/compensation.
7. Exact balanced unique-source GL, stock/party/cash/tax reconciliation, genuine openings/expenses/journals/reversals/period locks and no false no-restock COGS reversal.
8. All 12 report tabs, date presets/custom/as-of/event dates, roles/CSV, working-paper labels, A4/58/80 mm fields/reprints and independent sidebar scrolling/accessibility.
9. Auth/CSRF/MFA/rate limiting/operator injection/XSS/SSRF/upload, redacted errors/logs, append-only/non-disableable audit, legal holds/privacy rights and secret scanning/dependency checks.
10. Outbox overlap/lease expiry/stale-token/crash/backoff/dead-letter/provider reconciliation; private exports/download expiry and large-work budgets.
11. Atlas/TLS/pool limits, actual Vercel routes/HTTPS/env/runtime, protected previews, actual India-required logs/backups and isolated restore with number/provider/session reconciliation.
12. Migration preservation/cutover/rollback and professional policy/licensing/privacy/contract evidence. Correct known research limitations at actual release rather than hide them.

No release if mandatory isolation, money, stock, numbering, payment/return, GL, audit, legal eligibility or recoverability tests fail. A limited release is valid only for independently safe supported profiles with blocked unsupported functions.

12. Delivery, implementation order and your immediate response

Build incrementally: inspection/approved plan → CI/safe environments → tenancy/auth/field permissions → validators/indexes/exact money/legal policies → masters/purchases/stock with GL → atomic sales/printing → returns/refunds/corrections → complete accounting/reporting → durable audit/jobs/privacy/backup → approved integrations or tested exclusions → staging/migration/runtime/security/professional review → authorized production.

Deliver:

- Working JavaScript source and lockfile, versioned models/validators/indexes/migrations and APIs/data dictionary.
- Feature/permission/gap/requirement traceability, tests with actual raw results and a list of failures/limitations.
- Exact-money/rounding/accounting/legal-policy documentation and source/effective-date/reviewer register.
- `.env.example` placeholders only, Vercel/Atlas deployment steps, secure setup and operating guide.

- Migration/reconciliation/backup/restore/incident/runbook tools and evidence; no production secrets/DB/demo business data in archives.
- Owner/cashier/inventory/accountant instructions, complete reports/template documentation and a go/no-go record signed by the appropriate business/technical/professional-policy reviewers.

Your immediate first response must be the repository findings, approved-scope confirmation, gap analysis, unresolved factual/legal questions and phased implementation/testing plan. Do not claim the product is already built, deployed, tested or universally compliant. Do not start destructive changes or paid provisioning before approval.

08 Official and technical source register

Observed/referenced for this specification; research cut-off 26 October 2026.

This register is a reading/evidence index, not a statement that every amendment, applicability question or tenant configuration has been professionally approved. Living documentation changes. The legal matrix records failed retrievals, older compilations, future commencement and mandatory release checks. Follow the actual official instrument; do not infer law from a URL/title alone.

S01 — MongoDB documentation — transactions

https://www.mongodb.com/docs/drivers/node/current/crud/transactions/

S02 — Vercel documentation — function limits

https://vercel.com/docs/functions/limitations

S03 — CBIC — CGST Rule 49: bill of supply

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter6/rule49_v1.00.html

S04 — CBIC — CGST Rule 46: tax-invoice particulars

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter6/rule46_v1.00.html

S05 — GST Council/CBIC — Notification 78/2020: HSN detail

https://gstcouncil.gov.in/sites/default/files/2024-05/notfctn-78-central-tax-english-2020.pdf

S06 — CBIC — Notification 12/2017–CT(R), including veterinary healthcare entry

https://cbic-gst.gov.in/hindi/pdf/central-tax-rate/Notification12-CGST.pdf

S07 — CBIC — IGST Section 10: goods place of supply

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_IGST_Act/active/chapterv/section10_v1.00.html

S08 — GST Council — Circular 209/03/2024

https://gstcouncil.gov.in/node/4989

S09 — CBIC — CGST Section 15: value/discounts and amendment notes

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_CGST_act/active/chapter4/section15_v1.00.html

S10 — CBIC — CGST Section 34: credit/debit notes and amendment notes

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_CGST_act/active/chapter7/section34_v1.00.html

S11 — GST Council — e-invoice Notification 10/2023 listing

https://www.gstcouncil.gov.in/node/4365

S12 — GSTN — 30-day IRP reporting advisory

https://www.gst.gov.in/newsandupdates/read/543

S13 — GSTN — proposed recipient-generated RCM self-invoice advisory

https://www.gst.gov.in/newsandupdates/read/678

S14 — CBIC — CGST Rule 138: e-way information and conditions

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter16/rule138_v1.00.html

S15 — CBIC — CGST Section 16: input-credit conditions

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_CGST_act/active/chapter5/section16_v1.00.html

S16 — CBIC — CGST Rule 56: accounts, stock and electronic records

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter7/rule56_v1.00.html

S17 — CBIC — CGST Section 36: retention of accounts

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_CGST_act/active/chapter8/section36_v1.00.html

S18 — CBIC — CGST Rule 57: electronic backup and record production

https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter7/rule57_v1.00.html

S19 — Income-tax Department — notified Income-tax Rules 2026

https://www.incometax.gov.in/iec/foportal/sites/default/files/2026-03/En-Notified-IT-Rules-2026-20-03-2026.pdf

S20 — India Code — Companies accounts rules / section 128 reference

https://www.indiacode.nic.in/handle/123456789/1362/simple-search?query=Chapter+IX+-+The+Companies+%28Accounts%29+Rules%2C+2014.+&searchradio=rules

S21 — ICAI — revised 2024 accounting audit-trail implementation guide

https://eirc-icai.org/uploads/background_materials/Revised%202024_Implementation%20Guide%20on%20Reporting%20of%20Audit%20Trail%20(1)_1712114860.pdf

S22 — Income-tax Department — Act 2025, Section 186 (2026 page)

https://www.incometaxindia.gov.in/w/section-186-41

S23 — CDSCO — Drugs Rules 1945, retrieved 2022 compilation

https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2022/drug_rules/Drugs%20Rules,%201945%20(1).pdf

S24 — Central Bureau of Narcotics — official starting point for specialist review

https://narcoticsindia.nic.in/

S25 — Consumer Affairs — packaged-commodities compilation

https://consumeraffairs.gov.in/public/upload/admin/cmsfiles/whatsnews/Book_on_Legal_Metrology_Packaged_Commodities_Rules,2011_with_all_amendments_whatsnews.pdf

S26 — NPPA — DPCO source/amendment listing

https://nppa.gov.in/en/drugspricescontrolorder2013

S27 — NPPA — 30 June 2026 DPCO amendment, S.O. 3516(E)

https://nppa.gov.in/storage/uploads/pdf/dpco-amend-2026-57e460458980dab7263beda9399d68bc.pdf

S28 — CPCB — Bio-Medical Waste Management Rules 2016

https://cpcb.gov.in/uploads/Projects/Bio-Medical-Waste/Bio-medical_Waste_Management_Rules_2016.pdf

S29 — MeitY — DPDP Act commencement notification

https://www.meity.gov.in/static/uploads/2025/11/c56ceae6c383460ca69577428d36828b.pdf

S30 — MeitY — Digital Personal Data Protection Rules 2025

https://www.meity.gov.in/static/uploads/2025/11/53450e6e5dc0bfa85ebd78686cadad39.pdf

S31 — MeitY — December 2025 DPDP Rules corrigenda

https://www.meity.gov.in/static/uploads/2025/12/3c7ebbae0e5456f493f486e6845df86b.pdf

S32 — India Code — SPDI Rules 2011 primary-file reference (fetch/transition review pending)

https://www.indiacode.nic.in/ViewFileUploaded?path=AC_CEN_45_76_00001_200021_1517807324077%2Frulesindividualfile%2F&file=GSR313E_10511%281%29_0.pdf

S33 — CERT-In — directions dated 28 April 2022

https://www.cert-in.org.in/PDF/CERT-In_Directions_70B_28.04.2022.pdf

S34 — India Code — Bharatiya Sakshya Adhiniyam 2023

https://www.indiacode.nic.in/bitstream/123456789/20063/1/a2023-47.pdf

S35 — MCA — Companies (Accounts) Fourth Amendment Rules 2022

https://www.mca.gov.in/bin/ebook/dms/getdocument?doc=MTcyODIyOTI0&docCategory=Notifications&type=open

S36 — Vercel documentation — commercial-plan/fair-use conditions

https://vercel.com/docs/limits/fair-use-guidelines

S37 — MongoDB documentation — Atlas IP access lists

https://www.mongodb.com/docs/atlas/security/ip-access-list/

S38 — Vercel documentation — Cron management/authentication

https://vercel.com/docs/cron-jobs/manage-cron-jobs

Primary-PDF retrieval evidence

Selected primary PDFs were downloaded for textual research. Their hashes/pages/retrieval statuses are recorded in `research/primary-source-retrieval.json`; later reads and failures are recorded separately. Only selected excerpts are bundled, not the full PDF files. A failed download is not a reviewed full statute.